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Review article

Ontology learning towards expressiveness: A survey

Pauline Armary [a,b,*], Cheikh Brahim El-Vaigh [a], Ouassila Labbani Narsis [a], Christophe Nicolle [a]

[a] CIAD, UR 7533, Université de Bourgogne, UB, Dijon, F-21000, France
[b] Anabasis Assets, 32 Boulevard Ornano, Saint-Denis, 93200, France

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ABSTRACT

Ontology learning, particularly axiom learning, is a challenging task that focuses on building expressive and decidable ontologies. The literature proposes several research efforts aimed to resolve the complexities inherent in axiom and rule learning, which seeks to automatically infer logical constructs from diverse data sources. The goal of this paper is to conduct a comprehensive review of existing work in this domain. It aims to critically analyze the contributions and limitations of current approaches, providing a clear understanding of the state-of-the-art and identifying areas where further research is needed.

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* Corresponding author at: CIAD, UR 7533, Université de Bourgogne, UB, Dijon, F-21000, France.
E-mail address: pauline_armary@etu.u-bourgogne.fr (P. Armary).

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1. Introduction

Ontology, typically regarded as the conceptual representation of a specific domain [1], facilitates the creation of a comprehensive, semantically interpretable description of digital content for machines. Traditionally, constructing a domain-specific ontology is a manual endeavor undertaken by ontologists and knowledge engineers, to ensure accurate modeling adapted to the domain's specific needs [2]. This process requires a meticulous understanding of the domain in question. Furthermore, these experts incorporate domain-specific constraints into the ontology, essential for enabling logical inferences based on first-order logic, also known as deductive reasoning. The development of an ontology involves delineating the concepts or classes of entities pertinent to the domain and mapping their interrelations. The expressiveness of an ontology is contingent on the data model utilized, and the defined constraints enhance its capability for deductive reasoning. However, manual ontology construction is inherently labor-intensive and time-consuming. Consequently, Ontology Learning (OL) — the automated generation or refinement of ontologies from diverse information sources, including textual documents, databases, or insights from domain experts — seeks to streamline the ontology development and enrichment process [3,4].

Ontology learning covers all the techniques which help knowledge engineers to build or update an ontology of a given domain by automatizing its whole construction (or part of it), from different data sources. The ontology learning process starts with several sources of data, like databases or documents, and uses algorithms to extract components of the ontology (classes, relations, axioms, etc.) to integrate them into a new or existing ontology. Those components have been classified within the Ontology Learning Layer Cake proposed by [5] (see Fig. 1).

The OL Layer Cake begins with identifying and automatically learning domain-specific terms and their synonyms, representing the most straightforward layer. Beyond the fundamental learning of Concepts, Relations, and their Hierarchies, the framework encompasses the acquisition of Axiom Schemata, such as establishing a Disjoint constraint between two Concepts. It also facilitates, in certain instances, the derivation of General Axioms like Class Expression Subsumption. This involves defining rules by combining multiple concepts subsumed under another or employing a general first-order logic formula, as exemplified in the model's uppermost layer.

In the context of Semantic Web framework as established by the World Wide Web Consortium[1] (W3C), the initial five layers can be associated with the Resource Description Framework[2] (RDF) and RDF Schema[3] (RDFS) standard. These standards are used to define the classes, relations and their hierarchies in the knowledge graph, along with their corresponding instances. To provide more refined constraints over the data in the knowledge graph, the Web Ontology Language[4] (OWL) was introduced. OWL offers a syntax and semantics for defining axioms, which are restrictions over the classes and relations, as presented in the Axiom Schemata layer of the OL Layer Cake. The upper layer, General Axioms, sometimes named Rules, has been object to various propositions within the W3C to integrate rule language with OWL such as SWRL[5] or RIF.[6]

Therefore, ontology learning targets the different components of the T-Box of the Knowledge Base. Addressing the A-Box, that is the instances of the knowledge base, are covered by techniques often named "knowledge graph construction" or "automatic knowledge graph construction".

The ambition of ontology learning is to accelerate and automate the creation of ontologies, sometimes revealing hidden relationships in document corpora, while improving the integration in complex data environments. Ontology learning offers several advantages, including scalability in handling large-scale data and automation that reduces the need for manual construction. It also facilitates domain knowledge sharing with a common vocabulary shared across documents, and discovers new relations within the data that might not be apparent to experts [6–9].

However, ontology learning comes with limitations such as dependency on data quality, challenges in extracting complex relationships, significant computational resource requirements, and difficulty maintaining efficiency with very large datasets. Additionally, evaluating the ontology's quality can be complex, and differences in vocabulary between the ontology and human experts can lead to misunderstandings [6,7,10]. Ontology learning algorithms can, therefore, be used to accelerate the processing of heterogeneous data sources but still require human validation, which limits their total autonomy.

Since the grounding work in [11] from 2012, which provides an extensive survey of the Ontology Learning field from the beginning of the 2000s and its future perspective, several surveys have reviewed the progress of ontology learning. Most of these reviews are centered around a specific source of data, mostly from text [5–8,12–14], relational database [9,10,15–18]; or knowledge base [19]. Other surveys address the topic from the technical angle: Ontology learning tools have been extensively reviewed in two surveys [20,21]. Despite this comprehensive coverage, these reviews fail to address the tools' availability and developmental stage, as we note that many tools discussed by [20,21] are no longer accessible online. Additionally, three surveys review the emergence of deep learning within the field of ontology learning, analyzing the different algorithms used to address each learning stage of Ontology Learning Layer Cake [8,13,22]. However, there is often a lack of detailed descriptions regarding the specific axioms and rules employed in the referenced works.

The literature extensively discusses the initial five layers of the Ontology Learning Layer Cake, which contribute minimally to ontology enrichment, hence termed lightweight ontology learning [11]. Acquiring axioms (the two topmost layers of the OL Layer Cake), especially general ones, significantly enhances the ontology's expressiveness and reasoning capabilities. For instance, adding Disjoints, Symmetry, or Inverse axioms within the ontology enriches domain knowledge by allowing the imposition of more precise constraints on domain conceptualization. Our study explores how data sources can be transformed into formal ontological systems by adding those specific axioms to improve reasoning facilities.

Inspired by the systematic literature review methodology, our work examines approaches to heavyweight ontology learning, mainly focusing on the techniques applied to the model's topmost two layers. This

[1] https://www.w3.org/
[2] https://www.w3.org/TR/rdf12-concepts/
[3] https://www.w3.org/TR/rdf-schema/
[4] https://www.w3.org/TR/owl2-overview/
[5] https://www.w3.org/submissions/SWRL/
[6] https://www.w3.org/TR/rif-overview/

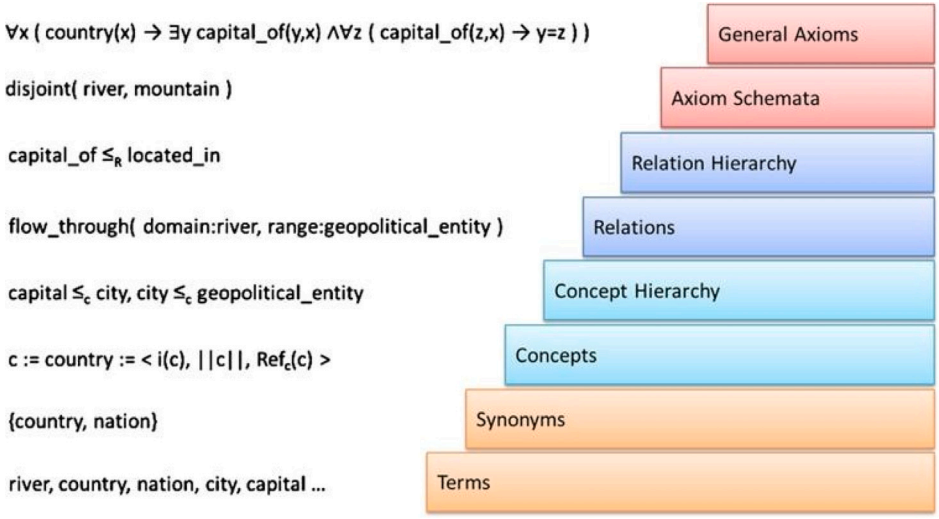


Fig. 1. The Ontology Learning Layer Cake defines the different elements composing the ontology, from the least to the most complex. *Source:* Fig. created by [5].

document is structured as follows. Section 2 describes the methodology used to conduct this literature review. Section 4 presents this review's results and analysis, while Section 5 shows a broad debate on ontology learning, showcasing different challenges and research directions. Finally, Section 6 presents a final overview of the findings of our review.

2. Review methodology

This survey methodology is inspired by the work in [23] which introduced the SLR methodology in the computer science field. It starts by identifying the key research question in Section 2.1, and then defines the review protocol used to conduct this review in Section 2.2. Finally, Section 2.3 presents the steps undertaken to perform the review.

2.1. Key research questions

The general focus of investigation on this survey is the following free-form question: *What can be done to build an expressive ontology automatically using ontology learning techniques?* OR *How can we transform a data source into a formal ontological system that enables reasoning?* This question can be analyzed as the convergence of three core focal elements: the data source, the target expressiveness (axioms and rules), and the algorithm. Therefore, we identified 3 main questions, each divided into sub-questions to cover the heart of our survey.

Ontology learning integrates diverse data sources like databases, web content, and documents into the ontology construction. It analyzes these sources to extract example instances or describe domain constraints in a logical format. This process relies on specific datasets, which may be publicly or privately assembled for distinct tasks or industrial use. The ontology's potential to enhance algorithm accuracy, such as entity linking tasks, is also leveraged. The data used is derived from a targeted application domain, influencing the type of ontology developed.

The research question **RQ1: What are they learning from?** aims to detail the data sources for learning axioms and rules, encompassing various sub-questions related to this topic, such as the data format addressed, the content conveyed by the data, the characteristics of the datasets used, the use of existing ontologies, and the application domain.

Ontology learning enhances expressiveness by targeting various axioms from the OWL standard and complex rules that combine multiple axioms. This approach primarily focuses on the top two layers of the ontology layer cake, analyzing the types of regulations and axioms addressed in academic papers. It employs a detailed classification based on the OWL2 axioms defined by the W3C. The logical formats for expressing these axioms and rules include OWL,[7] SWRL,[8] RIF,[9] SHACL,[10] Datalog,[11] Prolog, Description Logic, and First-Order Logic. The papers explore axioms with varying levels of expressiveness within Description Logics, denoted by mathematical letters like ELC and SHOIN (D). However, they sometimes may not capture the full expressiveness during the learning process. Including these axioms and rules enhance the ontology's inference capabilities, but at a trade-off with the knowledge base's decidability.

The purpose of **RQ2: What are they learning?** is to examine the targeted algorithms and include various related sub-questions about the axioms, rules, formal language, and level of expressiveness targeted.

Different algorithms from the field of Artificial Intelligence are used to process diverse data sources and achieve targeted expressiveness. It includes rule-based algorithms, machine learning with unsupervised statistical techniques, and artificial neural networks. Learning algorithms requires a defined evaluation protocol of the algorithms' capabilities and the quality and efficiency of the knowledge extracted. Furthermore, learning algorithms usually utilize existing technologies and technical frameworks to implement and test their solutions, such as specific programming languages and libraries. To promote community engagement and reuse of their findings, the solutions may be implemented as tools or standalone scripts, with some being open-source and others available as SaaS solutions, facilitating collaboration and knowledge sharing.

The question **RQ3: How do they learn?** focuses on the process used to extract axioms from data. The question is linked to sub-themes such as the type of algorithm used, the type of evaluation carried out, the technical framework developed, and the accessibility to the community of the implementation.

2.2. Defining the review protocol

The SLR protocol review starts by identifying the keywords and the appropriate databases that allow us to select the relevant article

[7] https://www.w3.org/TR/owl2-overview/
[8] https://www.w3.org/submissions/SWRL/
[9] https://www.w3.org/TR/rif-overview/
[10] https://www.w3.org/TR/shacl/
[11] https://datalog.dev/

in Section 2.2.1. Next, it explains the exclusion/inclusion criteria for selecting articles from these databases in Section 2.2.2.

2.2.1. Databases selection

This study uses five scientific databases: Web of Science, ScienceDirect, ACM Digital Library, IEEE Explore, and arXiv. These databases, which satisfy the SLR reproducibility criteria and support complex search queries such as Boolean aggregators (e.g., AND, OR) [24], are selected for this study. In contrast, databases like Google Scholar or Semantic Scholar were disregarded due to their inability to meet the criteria above.

The query is formulated based on the study's two primary keywords: *Ontology Learning* and *Rule*. Conjunctions and disjunctions are used to include variations of these terms, such as *Axiom*, *description logic* and *OWL DL* for *Rule* or *knowledge base learning*, *schema induction* and *schemas learning* for *Ontology Learning*. These variations are identified through comprehensive reviews in the field of Ontology Learning [6–8,12–14], and by analyzing keywords manually extracted from relevant literature. The final query is expressed as follows:

> "ontology learning" OR "automatic ontology construction" OR "ontology enrichment" OR "ontology extension" OR "knowledge base learning"
> AND
> "rule" OR "axiom" OR "SWRL" OR "description logic" OR "OWL DL" OR "schema induction" OR "axiom extraction" OR "schemas learning" OR "DL-learning".

2.2.2. Exclusion criteria

A set of exclusion criteria helps refine the articles retrieved from the keyword search and retain pertinent papers that address our research questions. In our survey, we identified the following nine exclusion criteria:

ExC1: Papers published before 2012

ExC2: Duplicate papers returned by the various search engines

ExC3: Papers not written in English

ExC4: Summary of Conference or Workshop which presented several works

ExC5: Paper that was not peer-reviewed and published

ExC6: Papers which were not accessible online

ExC7: Survey or theoretical papers with no experiment

ExC8: Papers that presented methodological considerations on ontology building without automatic learning

ExC9: Papers which describe a light-weight ontology

To address exclusion criteria ExC9 precisely and limit this survey to the most expressive ontology, the selection was restricted to the highest two levels of the Ontology Learning Layer Cake. Therefore, all papers that target concepts, conceptual taxonomy, and relations were excluded based on the general description and the axioms mentioned in the paper. A refined analysis was performed on articles mentioning the subclass axiom or concept inclusion axiom because this axiom is sometimes used in the literature to capture more complex information, such as complex class expression subclass, which is equivalent to extended definitions or complex implications. The following distinction was applied to select the papers: articles that mention **Subclass axioms concerning only classes** were classified as simple taxonomy and were not selected (ExC.9, no expressive ontology). Articles that mention **Subclass axioms with more complex information**, like Class Expression or other axioms, were classified as Class Expression or Class Axiom and were kept in the selection. Most articles that mention "General Concept Inclusion" are primarily concerned with this selection.

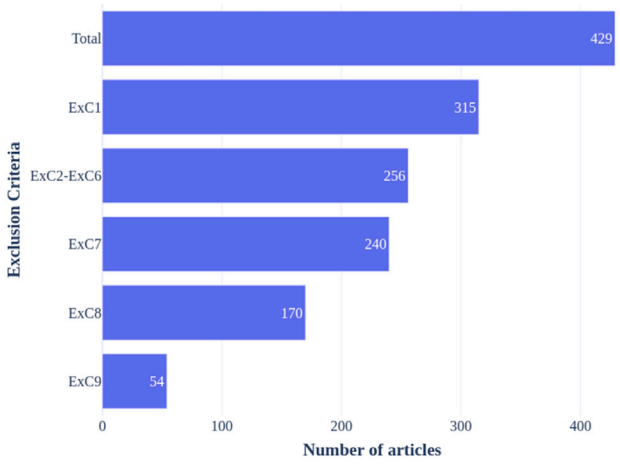


Fig. 2. Selection of articles with exclusion criteria.

2.3. Performing the review

The query described above is utilized to retrieve literature articles from databases, as outlined in Section 2.3.1. Subsequently, criteria were applied to these articles, retaining only those that met the specified requirements in Section 2.3.2.

2.3.1. Collecting articles

The query was performed on May 2023 via an automatic Python script to connect to the publisher's API (supported by the findpapers Python library [25] or via a manual query on the publisher's website when the API key was not available (for IEEE and Web of Science).

2.3.2. Applying exclusion criteria

The query gave a result of 429 unique articles after removing duplicates from the different databases. Articles anterior to 2012 were filtered to focus on the most recent papers from the last 10 years. The year 2012 also corresponds to the publication of the survey "Ontology Learning from text: A Look Back and into the Future" [11], which gave a clear overview of the state-of-the-art Ontology Learning for the research done during the first decade of the century.

The steps for applying the exclusion criteria and their results on the number of papers are presented in Fig. 2. Exclusion criteria 2 to 6 were performed within our retrieved table of retrieved papers to eliminate duplicated works, references to multiple works (within proceedings, for example), and papers that were not peer-reviewed and published, were not written in English, or were not accessible. In the second step, the title and abstract of each article were read to eliminate surveys and theoretical papers providing an overview of the task with no detailed implementation (ExC7). The obtained 240 articles were read to identify if they fall under the last two exclusion criteria (ExC8-9), targeting an expressive ontology with automatic methods. 70 articles were excluded as they did not provide an automatic method of learning or did not target an ontology (no ontology learning), and 116 were excluded as they did not target axioms or rules but mostly taxonomies or non-hierarchical relations. As a result, 54 articles were selected within the scope of our review.

3. Overview of included studies

This section analyzes the statistic distribution of all included studies before analyzing the papers along our three main research questions. It reviews the evolution of expressive ontology learning and its geographic distribution.

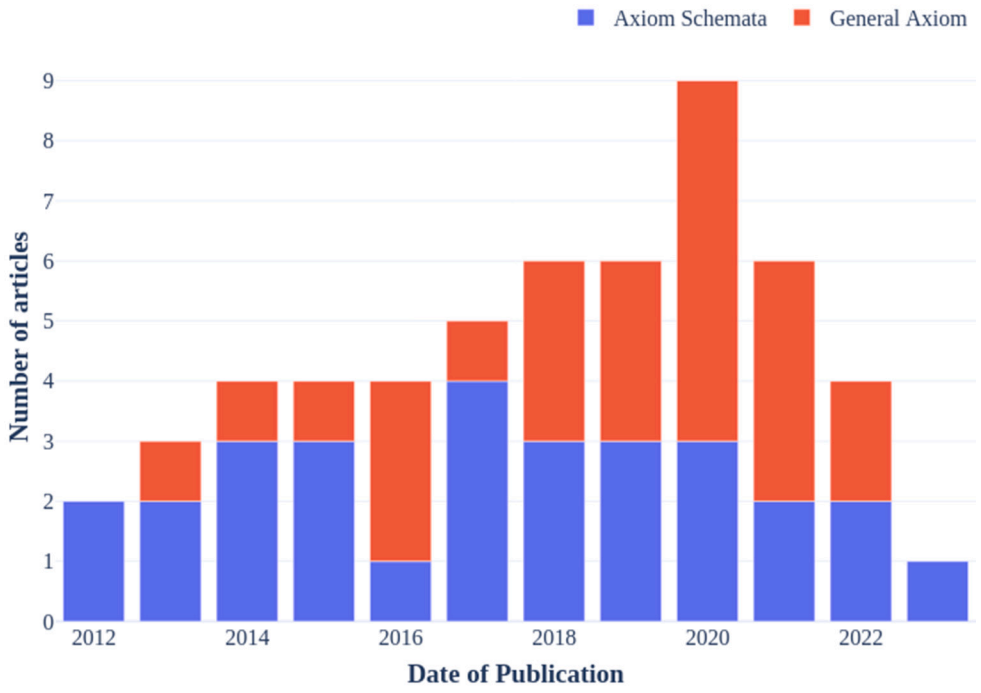


Fig. 3. Number of papers per year learning the last two levels of the ontology learning layer cake: axiom schemata and general axiom.

To understand the evolution of expressive ontology learning in the last decade, Fig. 3 plots the number of papers per year, highlighting the evolution according to the two top levels of the OL layer cake addressed in the study: axiom schemata and general axiom. The field is slightly increasing over time, with a clear growth of interest towards general axiom, the highest level of the layer cake. Despite an exceptional number of papers for the year 2020, the general tendency seems to be comforting the idea that recent years tackled the issue identified in [11] to go from light-weight ontology to more expressive (heavy-weight) ontologies.

To understand the geographic distribution of the main contributors of the field, Fig. 4 plots the number of papers per country. The most represented continents are Europe, Asia, and America. Europe is the main continent contributing to the field, with almost half of the articles coming from one of its countries. France is the main country and contributes alone to 15 percent of the papers of this study. However, the multiplicity of contributing countries (France, Germany, Italy, Poland, United Kingdom, Spain, and Hungary) allows Europe to come on top of the contribution. In Asia, China and India are the main contributors. In America, Brazil, Mexico, and Canada provide the main studies. Surprisingly, the United States contributes very little to this study, with only one article, despite being the leading country in terms of R&D development.[12]

This lead of Europe and the almost absence of the United States may be explained by this study's strong focus on the expressiveness of ontology and the highest levels of layer cake. This focus highlights studies heavily influenced by Logic, which comes from a long tradition in Europe.

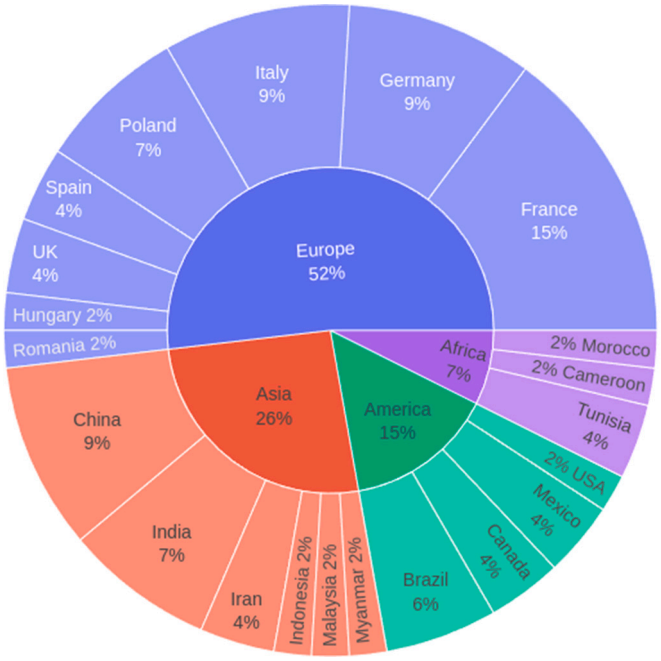


Fig. 4. Number of papers per country of the main author.

4. Results and analysis of the review

This section provides a detailed analysis of the survey results over the three axes defined in the Research Question section: *RQ1.What are they learning from?*, *RQ2.What are they learning?*, *RQ3.How do they learn?*. It analyzes the papers selected after the application of all exclusion criteria.

4.1. What are the articles learning from? (RQ1)

Ontology Learning relies on data integrated as source of information, databases, web content, documents and the like. The different sources of data are a key component of the ontology learning process, as they constrain the type of treatment to undertake based on how structured or unstructured the format and content of information is provided. This section addresses the question, *What are the literature articles learning from?*. This question is divided into four subquestions:

[12] https://data.oecd.org/fr/rd/depenses-interieures-brutes-de-r-d.htm

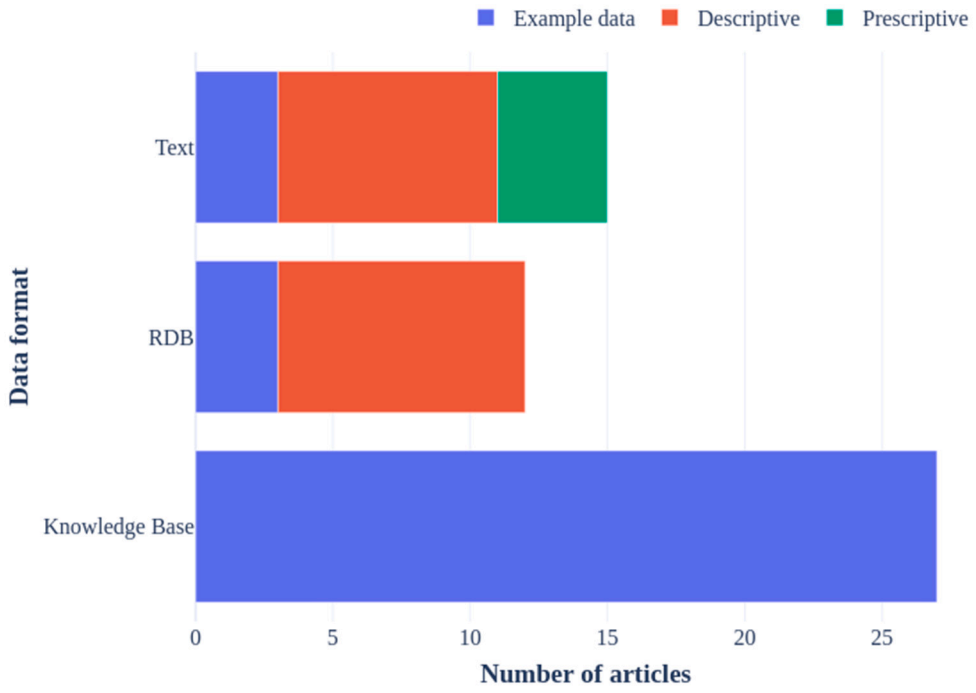


Fig. 5. Distribution of the number of articles with respect to data format and content (Answer to RQ1.1).

RQ1.1: what is the data format of the source?; **RQ1.2**: what is the type of content conveyed by the source?; **RQ1.3**: which specific dataset are used?; **RQ1.4**: is there an initial ontology the paper relies on during the ontology learning process?; **RQ1.5**: what is the application domain of the data source?

Our literature review identified four main data types utilized in Ontology Learning in Section 4.1.1. Section 4.1.2 offers to classify the content of these data sources into three categories. Details on OL datasets, OL existing ontologies, and application domains are provided in Sections 4.1.3, 4.1.4, and 4.1.5, respectively.

4.1.1. Data format (RQ1.1)

In the literature, ontology learning is performed with unstructured data (Text), semi-structured data (XML), and structured data (RDBs and KBs).

Most literature in the field utilizes **Text documents**, **relational databases (RDBs)**, and **knowledge bases (KBs)** such as RDF format for learning. Only one article was identified that learns from XML semi-structured format, as a format of extraction of relational database and was therefore associated with RDB format.

The distribution of the different articles based on data format is shown in Fig. 5, with a joint analysis along the content format presented in Section 4.1.2.

Knowledge bases and Text documents are the two predominant data formats employed in ontology learning, often through an inductive learning approach, which infers general rules from specific examples. In ontology learning, this approach may involve analyzing textual documents, structured data, or other sources to discern patterns that illustrate examples, concepts, and their interconnections within a domain. Meanwhile, RDBs are frequently used where the constraints of the database are converted into axioms for a knowledge base. The process of translating data from an RDB to a KB entails mapping the schema, tables, and data instances from the relational database into concepts, properties, and individuals in an ontology, thereby transforming the content of an RDB into a semantically enriched format.

Overall, we can identify two different tasks in the literature of OL: **a Translation task** (a term proposed by [26]), which consists in transforming the information stored in one data format into another one, either from a constrained language to another (RDB to KB), or from natural language to a constrained language, where the challenge consists on solving the ambiguity of the language (from unstructured text to KB); **an Inductive task**, which consists in inferring causal relations based on the statistical observation of a great number of correlations.

4.1.2. Data content (RQ1.2)

Our review proposes to organize the content of the different data sources into three categories in the literature: example data, descriptive data, and prescriptive data. **Example data** consists of specific instances or examples that illustrate concepts within a domain. This type of data provides concrete examples of how concepts are instantiated or used in practice. It corresponds to the instances of an A-Box or the content of RDB tables. **Descriptive data** provides the description of the characteristics, properties, and relationships of entities within a domain in a general and abstract manner. Finally, **Prescriptive data** (or normative data) is a kind of descriptive data originating from documents that have a certain role in the discourse field as they enforce a normative state by their existence and their mode of publication. Those documents state how the domain should be and give precisely its constrains. Therefore, they are a more powerful source of knowledge since they tell in a precise and semi-structured manner the constraints of the domain (legal documents, regulations, code of conduct, etc.). They correspond to the performative mode of discourse as first presented by J.L. Austin in [27].

Fig. 5 shows the type of data for the different data formats. All the articles learning from the knowledge bases exploit the example data, whereas it is mostly the descriptive data that is used in the cases of text and RDB. We also identified some prescriptive data in text documents, mostly from clinical guidelines [28,29] or domain regulations [30,31]. We also surprisingly found that legal documents such as country laws are not directly addressed by the ontology learning community targeting expressive ontologies.

One may consider data format and data content redundant since learning from example data seems to be only from KBs, and description data would come from other sources (text documents and RDBs). However, we identified within the literature approaches that build example data from text documents and then use those examples to

Table 1
Data source: datatypes and content types (Answer to RQ1.1 and RQ1.2).

Data types	Content type	Articles
Knowledge base	Example data	[34–60]
RDB	Descriptive document	[61–69]
	Example data	[70–72]
Text	Descriptive document	[26,29,73–79]
	Example data	[80,81]
	Prescriptive	[28,30,31,82]

identify the axioms using techniques like named entity recognition [32] and relationship extraction [33]. The summary of articles for each data format and type is presented in Table 1. This summary shows that example data and descriptive data are utilized in both text documents and relational databases (RDB), whereas prescriptive data is exclusive to text documents. Knowledge bases (KBs) solely provide example data.

4.1.3. Datasets (RQ1.3)

Ontology learning involves the creation of complex and detailed ontologies typically from large-scale, heterogeneous datasets. These datasets often contain diverse types of information, including text documents, structured data, and various forms of knowledge representations. They encompass diverse domains and sources, each tailored to specific research objectives and methodologies. The following categories represent key types of datasets found by this survey:

Encyclopedic Knowledge Bases: Exemplified by resources such as DBPedia[13] e.g. [29,34–37,60,70,77,79], YAGO[14] [34,43,48] or Wikidata[15] [34,48], its offer comprehensive repositories of structured knowledge spanning a wide range of domains. These datasets consolidate facts and serve as foundational resources for semantic interoperability, information retrieval, and natural language processing research.

Repository Datasets: It is made of structured collections of data typically stored within databases or data repositories and used for training ontology learning algorithms. These datasets may cover diverse domains and applications, ranging from scientific data repositories to cultural heritage archives. NorthWind [63,66], Foursquare dataset [71] and University dataset [61],

Domain-Specific Ontologies: Domain-specific ontologies, such as the Semantic Sensor Network (SSN) Ontology, BioPAX, Wine ontology, Semantic Bible, Proton Ontology, and VSAO (Vertebrate Skeletal Anatomy Ontology), are specialized ontologies tailored to specific domains or application area including healthcare, sensor networks, and cultural heritage. These ontologies encode domain concepts, relationships, and constraints and are used in the literature to build example data for OL [26,43,52,54,55,59,74,77]

Textual Datasets: It contains content extracted from sources such as web pages, scientific articles, and guidelines documents. These datasets include general web extraction datasets, NEL (named entity linking) datasets, Wikipedia dumps [79], and guideline repositories. Textual datasets provide rich sources of linguistic and contextual information for natural language processing, text mining, and information retrieval tasks to build or enrich an ontology [28–30,57,73,75,79–81].

Private Companies Datasets: Private companies datasets comprise proprietary data sources maintained by commercial entities for internal use or commercial purposes. While access to private companies' datasets may be restricted, they represent valuable sources of real-world data for applied research and industry collaborations. The investigation revealed the Thales avionics maintenance ontology [46] and the Airbus Guidelines dataset, which consists of 960 system installation design principles [31].

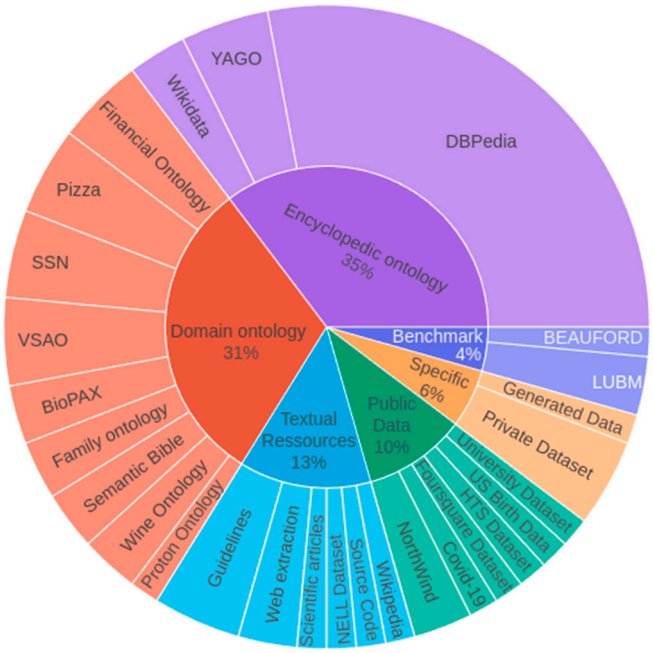


Fig. 6. Distribution of ontology learning datasets since 2012. (Answer to RQ1.3).

Benchmarks: The investigation identifies two benchmarks in ontology learning literature namely LUMB[16] (Lehigh University Benchmark) [54,59] and BEAUFORD (Benchmark for Evaluation of Ontology Alignment Algorithms) [74], which serve as standardized datasets and evaluation frameworks for assessing the performance of OL algorithms. Benchmarks typically consist of synthetic or real-world data designed to evaluate specific aspects of a research task, such as ontology alignment, entity matching, or semantic similarity.

The investigation revealed no mention of general ontologies or upper ontologies in the literature, despite their significance alongside domain ontologies as important subsets of ontologies. This may be due to the precise definition of those ontologies, which are limited in scope and often do not have instances associated with their classes and relations (A-box). They serve as reference guides for domain ontologies and, therefore, are not appropriate as a source dataset to learn new axioms. They may be used as guides during the learning process, as analyzed in Section 4.1.4 above.

The different types of datasets are grouped in Fig. 6 in two levels of classification, this figure shows particularly the Encyclopedic Knowledge Bases to be mainly used in the literature. Finally, Table 2 gives a straightforward view of the literature articles with respect to the different datasets mentioned above.

4.1.4. Existing ontology (RQ1.4)

The approaches employed in ontology learning often rely on foundational ontologies, which serve as guiding frameworks for the entire process. Here's an overview of the existing ontologies used in ontology learning:

Semantic structure Ontology: The investigation found two ontologies of semantic structure used in the papers reviewed. WordNet — a lexical database that organizes words into sets of synonyms, called synsets, and describes semantic relationships between these words

[13] https://www.dbpedia.org/
[14] https://yago-knowledge.org/
[15] https://www.wikidata.org/wiki/Wikidata:Main_Page
[16] https://swat.cse.lehigh.edu/projects/lubm/

Table 2
Dataset and their corresponding articles (Answer to RQ1.3).

Dataset type	Dataset	Article-reference
Benchmark	BEAUFORD	[74]
	LUBM	[54,59]
Domain ontology	Pizza	[26,74,77]
	SSN	[26,74,77]
	VSAO	[26,74,77]
	Family ontology	[54,59]
	Semantic Bible	[54,59]
	BioPAX	[43,52]
	Financial Ontology	[43,52,59]
	Wine Ontology	[54,66]
	Proton Ontology	[55]
Encyclopedic ontology	DBPedia	[34,36–41,43–45,47,48,50,51,55,56,58,60,83]
	YAGO	[34,43,48]
	Wikidata	[34,48]
Public Repository	NorthWind	[63,66]
	University Dataset	[61]
	Foursquare dataset	[71]
	Covid-19	[70]
	US Birth data	[70]
	HTS Dataset	[53]
Specific Data	Private Dataset	[31,46,72]
	Generated Data	[35]
Textual Resources	Guidelines	[28–30]
	NELL Dataset	[57]
	Wikipedia	[79]
	Scientific Articles	[80]
	Source code	[75]
	Web extraction	[73,81]

— and PPDB a specialized ontology focused on the properties and characteristics of pesticides. These ontologies are helpful in the process of ontology learning since they allow the grouping of terms based on their semantic relationships in the ontologies [28,29,74,76,78,79,82].

Domain Ontology: Domain ontologies are used to build datasets for ontology learning, as motioned above. Moreover, they are valuable resources to identify entities in the text. Our investigation found several domain ontologies, such as PRM Ontology for management documents analysis, BioPortal, SNOMED, and Disease Ontology in healthcare and biomedical textual documents analysis [30,75,80].

4.1.5. Domain of application (RQ1.5)

This section describes ontology learning application domains. Fig. 7 illustrates that over 63% of the studies do not exclusively target a single application domain. Instead, authors often address specific challenges by leveraging generalist or adaptable datasets, promoting the re-usability of their methodologies across diverse application domains. Notably, the industrial sector emerges as the most prevalent application domain, accounting for 20% of the studies, attributable to the development of domain-specific ontologies tailored for industrial contexts. Furthermore, a considerable portion of application domains is addressed by only 1% of the studies(in Communication, Finance, Computer Science, Risk Management, Teaching or Religion), underscoring the widespread adoption of ontology learning frameworks throughout the literature.

Overall, Section 4.1 delves into various facets of the data sources employed across the surveyed literature. In Section 4.1.1, four distinct data formats are identified: databases, XML, knowledge bases, and

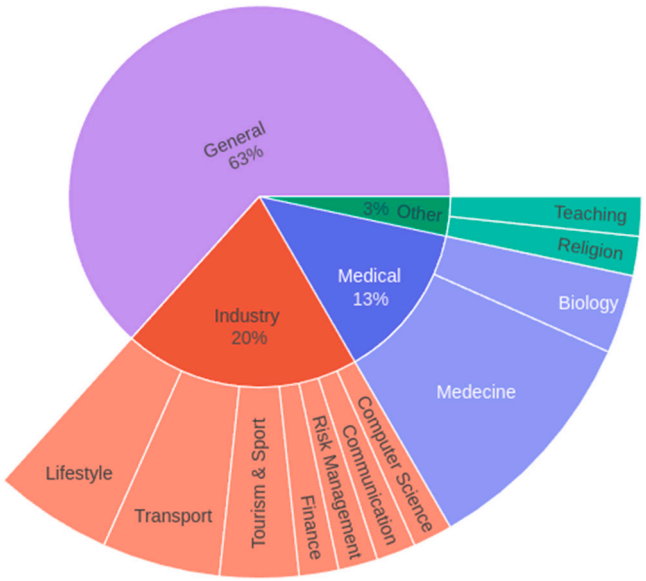


Fig. 7. The distribution of articles by domains (Answer to RQ1.5).

textual documents. Notably, knowledge bases emerge as the preferred format in the selected literature for learning heavyweight ontologies due to their high expressiveness. Furthermore, Section 4.1.2 categorizes data into two main types: example data, showcasing domain instances, and descriptive data, providing domain descriptions in either formal or informal terms. A subset of descriptive data, prescriptive data, is highlighted for its regulatory statements. This section also delineates two ontology learning tasks: inductive and translation tasks. Moving forward, Section 4.1.3 presents specific datasets classified into six categories: benchmark, domain ontology, encyclopedic ontology, public repository, specific data, and textual resources. In Section 4.1.4, two types of ontology utilized in the literature are outlined: semantic ontology, which describes word structures, and domain ontology, guiding vocabulary selection, particularly for textual documents. Lastly, Section 4.1.5 shows the broad application domains leveraging ontology learning within the literature. In conclusion, the response to *RQ1 (What are the literature articles learning from?)* furnishes a comprehensive understanding of the diverse data sources, formats, content nature, datasets, ontology types, and application domains encompassed within the field of ontology learning.

4.2. What are they learning? (RQ2)

Ontology learning enhances expressiveness by targeting various axioms from the OWL standard and complex rules that combine multiple axioms. This section focuses on the top two layers of the ontology layer cake, our review aims to answer the RQ2: *What are the literature articles learning?* In other words, what are the common learned axioms and rules in the literature?

The objective of this analysis is to define the level of expressiveness achieved by ontology learning literature towards an expressive or heavyweight ontology. To cover the different aspects of this question, this section will analyze three different subquestions: **RQ2.1**: what are the kind of axioms and rules targeted by the literature articles? **RQ2.2**: what are the targeted languages used to express those axioms and rules? **RQ2.3**: what are the levels of expressiveness explicitly targeted by the paper? **RQ2.4**: what are their objective for learning expressive axioms from the data?

Section 4.2.1 presents a detailed classification based on the OWL2 axioms defined by the W3C and analyzes the distribution of those

Table 3
Correspondence between the levels of the Ontology Learning Layer Cake, the W3C classification, and the axioms of Description Logic (Answer to RQ2.1).

OL layer cake	OWL classification	Axioms
A-Box	Class assertion, role assertion	instanceOf
Terms	Labels	Annotations, SKOS
Concepts	classes, class expression	complement, intersection, union, existential and universal quantification, cardinality
Concept hierarchy	class inclusion, General Class Inclusion (GCI)	subclass, equivalence
Relations	relations	relations
Relation hierarchy	property inclusion	property inclusion, property equivalence
Axiom Schemata	Class axioms	disjoint classes
	Object property axioms	transitivity, property chain, reflexivity, irreflexivity, symmetry, asymmetry, disjoint property, functional property, inverse functional property
	Data property axioms	disjoint data properties, data property domain and range, functional data property
General Axiom	GCI with class expression	Subsumption + complement, intersection, union, existential and universal quantification, cardinality
	–	Complex rules

axioms in the literature. Section 4.2.2 reviews the logical formats for expressing these axioms and rules with the W3C standards and other logical languages. Section 4.2.3 analyzes the level of expressiveness within Description Logic as described explicitly in the literature. Finally, Section 4.2.4 presents the objectives of the papers reviewed to learn expressive axioms.

4.2.1. Classification of axioms and rules (RQ2.1)

This survey focuses on the two upper levels of the layer cake (Axiom schemata and General Axioms). We base our classification of the axioms and rules on the standard of the W3C for the OWL2[17] which is the most expressive data model.

Table 3 presents the correspondences between the W3C classification, with the different classes of axioms defined in the standard, the name of the axiom associated to each class and the Ontology Learning Layer Cake levels as proposed by [84]. Moreover, this table gives a more fine-grained classification of the last two levels of the layer cake, axiom schemata, and general axiom. Axiom Schemata can be divided into:

Class expressions are the different forms of composed class: negation, conjunction, disjunction, existential and universal restriction and cardinality;

Class axioms defines the constraints on the class: concept inclusion, equivalence and disjointness;

Object property axioms defines the constraints on the object properties: transitivity, property inclusion, property chain, reflexivity, irreflexivity, symmetry, asymmetry, disjoint property, inverse property, functional property, and inverse functional property.

Data property axioms defines the constraints on the data properties: datatypes, data subproperties, disjointness, functional data properties.

Several works were identified that do not adhere to the axiom terminology prescribed by the W3C standard and Description Logic vocabulary, instead using the term "rules" [28]. Since this terminology was ambiguous to match with the level of expressiveness targeted, especially when the constraints are expressed in formal language outside the OWL Standard or Description Logic terminology, our analysis attempted to compare them with the axiom classification defined above. When those "rules" did not correspond to one single schema of axiom but required a combination of several axioms (especially with a subsumption axiom and other class and object property axioms), the paper was labeled as "complex rules", which correspond to the highest level of the ontology learning layer cake, "General axiom". As an example, [28] presents the following "rule":

$$\begin{aligned} ADcase = & cvdo : hasSymptoms(?p, cvdo : Douleur) \wedge \\ & cvdo : hasSymptomsValue(?s, \\ & \text{“}douleur_thoracique_aigue\text{”}) \wedge \\ & cvdo : Patient(?p) \wedge \\ & cvdo : hasSymptoms(?p, cvdo : Hypertension) \wedge \\ & cvdo : HasSymptomsValue(?s, 14) \\ & \rightarrow cvdo : RecommandedDiagnosis(?p, cvdo : Angioscanner) \end{aligned}$$

We further provide a fine-grained classification to the "General Axiom" category as follows:

Class Expression Subsumption, sometimes presented as General Concept Inclusion with Class Expression axioms, corresponds to articles learning a general definition of a class as they are composed of a subsumption axiom between a class and a class expression. The objective of those papers is to define the precise description of a class to assert the necessary (all individuals within the class will have the corresponding properties: in the form $C(x) \sqsubseteq (D(x) \wedge E(x))$ and sufficient (all individuals with the set of properties will belong to the class: in the form $(D(x) \wedge E(x)) \sqsubseteq C(x)$) conditions of the class attribution. This category is then redivided by the type of component learned within the class expression (negation, class conjunction, class disjunction, existential and universal restriction, and cardinality restriction).

Complex Rules are found in articles learning a combination of different axioms, in particular subclass axioms with a conjunction of class expression and relations, which are asserted not only to a class but also to another conjunction of class expression and relations. This category corresponds most often to articles that are not expressed with OWL or DL axioms but with more expressive formal languages for "rules" (SWRL, OWL-RL, FOL, etc.) as described in Section 4.2.2.

Fig. 8 depicts the proportion of each kind of axiom learned within the surveyed articles. Since two-thirds of the articles (40 over 54 articles) target two axioms or more (ranging from 1 to 12), this distribution does not correspond to the proportion of articles but the proportion of axioms over all the axioms learned in the surveyed literature.

This diagram conveys an overall homogeneity within the targeted axioms, with almost 1/4 of articles targeting each type. The only family of axioms that is poorly addressed in the articles is the Data Property axioms, with only four articles [49,50,63,69]. A greater disparity is notable among the different axioms targeted. Subsumption axioms are learned by a great number of articles [26,37–39,42,43,46,58,59,62–66,70,74,77,80,83], which comes with no surprise since they are a key aspect of General Concept Inclusion with class expression as well as complex rules. The other well-addressed axioms within the

[17] https://www.w3.org/TR/owl2-syntax/

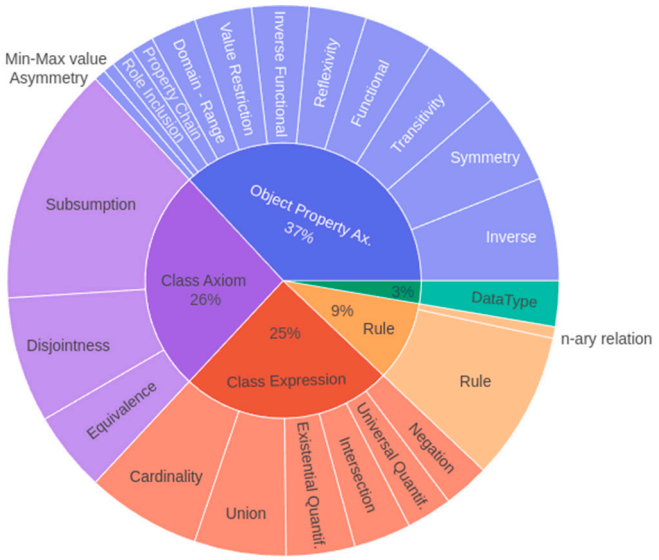


Fig. 8. Number of axioms by type learned within the articles. Since articles often extract several axioms at once (transitivity and symmetry), this distribution does not correspond to the number of articles but the number of mentioned axioms within the literature. (Answer to RQ2.1).

surveyed articles are cardinality [39,61–63,65,66,68,69,74,77], disjointness [29,41,44,45,47,55,59,63,66,73], union [30,38,45,46,58,66,79,80], inverse property [31,51,56,60,62,65–67,81] and symmetry [35,40,51,60,61,64,66,81]. On the other hand, role inclusion [57,60], property chain [36,66], asymmetry [66] and min–max value [50] are scarcely covered in the reviewed literature.

Table 4 provides a detailed view of the literature articles with respect to the different axioms mentioned above.

Figs. 9 and 10 present a comparative analysis[18] of the learned axioms with respect to data format (database, knowledge base, text), as defined in Section 4.1.1 and content (example, descriptive and prescriptive data), as defined in Section 4.1.2.

Fig. 9 presents a certain homogeneity: all types of axioms and rules are learned from the different data formats: there is no clear privilege of one format versus another to target some axioms. However, there are some visible preferences: **Relational Database** (in blue) are mainly focused on identifying object property axioms [61,62,64–68]. This makes sense since the most important information to capture from the relational schema is the constraints over the relation between tables. Some papers also considered data property [69], class expression subsumption [63] and complex rules [70–72] **Text document** (in green) is used mainly to learn complex rules [28,30,75,76,78,82], then class axioms [29,73,74,77] and class expression subsumption [26,79,80]. **Knowledge Base** (in red) are the most diverse source of axioms, addressing mostly class expression subsumption [37–39,42,43,46,54,58], then class axioms [41,44,45,47,55,59], object property axioms [35,36,40,51,56,57,60], data property axioms [49,50] and complex rules [34,48,52,53].

The contrast is more pronounced in Fig. 10 when considering the content of the source data as defined in Section 4.1.2: **Example data** (in blue) is used to learn all kinds of axioms via induction: class axioms [41,44,45,47,55,59], object property axioms [35,36,40,51,56,57,60,66,81], data property axioms [49,50], class expression subsumption [37–39,42,43,46,54,58,80] and complex rules [34,48,52,53,70–72]. **Descriptive data** (in red) is also used to identify all kinds of axioms. Further analysis establishes the distinction between database schema, which is used to identify object property axioms [61,62,64–68] and data property [63,69], whereas text document focuses on

[18] The Sankey diagram used in both figures allows us to define the distribution of a value towards a second target value. It is read from left to right, with the source of information on the left and how it is distributed to the target on the right. Therefore, those two Sankey diagrams should be read as "n% of the articles learning from knowledge base are targeting class expression subsumption".

Table 4
Articles depending on the kind of axioms and rules they learn (Answer to RQ2.1).

OWL axiom family	Axiom	Articles
Class Axiom	Subsumption	[26,37–39,42,43,46,58,59,62–66,70,74,77,80,83]
	Disjointness	[29,41,44,45,47,55,59,63,66,73]
	Equivalence	[29,49,63,65,72–74]
Class Expression	Cardinality	[39,61–63,65,66,68,69,74,77]
	Negation	[31,42,54,79]
	Universal Quantification	[41,48,69,77]
	Union	[30,38,45,46,58,66,79,80]
	Intersection	[38,45,50,79,80]
	Existential Quantification	[37,41,44,58,69,77]
Data Property Axiom	DataType	[49,50,63,69]
Object Property Axiom	Inverse	[31,51,56,60,62,65–67,81]
	Symmetry	[35,40,51,60,61,64,66,81]
	Transitivity	[35,36,40,60,64,66,81]
	Asymmetry	[66]
	Value Restriction	[37,41,44,63,74]
	Reflexivity	[35,40,60,66,81]
	Domain - Range	[43,63,65,69]
	Role Inclusion	[57,60]
	Property Chain	[36,66]
	Min-Max value	[50]
	Functional	[60,61,65,66,69,81]
	Inverse Functional	[60,61,66,68,81]
Rule	Rule	[28,30,34,48,52,53,70–72,75,76,78,82]
	n-ary relation	[67]

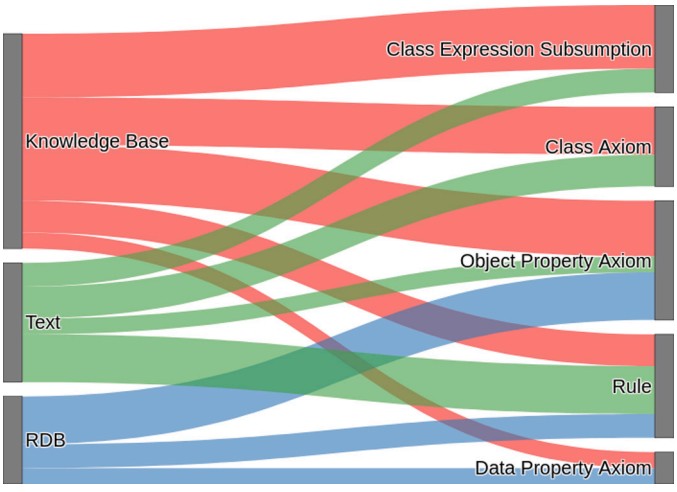


Fig. 9. The articles learn different axioms based on the format of the data source using Sankey diagram . A connection between a data format (e.g. Knowledge, based) and an axiom (e.g. Class, Axiom) means this data is being used to learn this axiom. The size of the link indicates the number of articles (Knowledge Base is used for Class Axiom more than Rule). (Answer to RQ2.1).

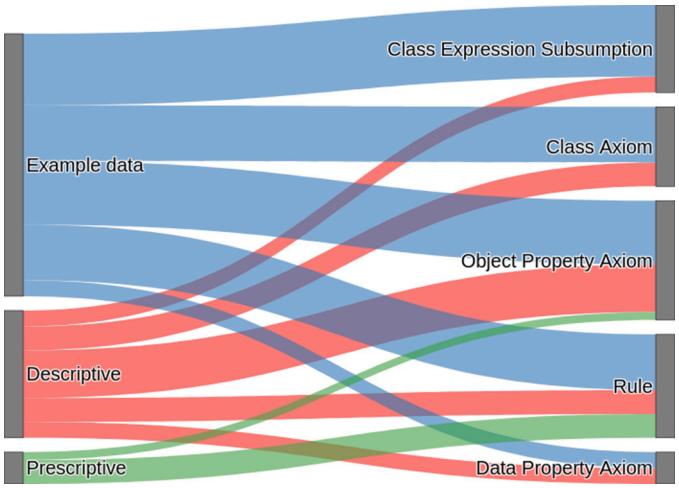


Fig. 10. The articles learning different axioms based on the content of the data source (Answer to RQ2.1).

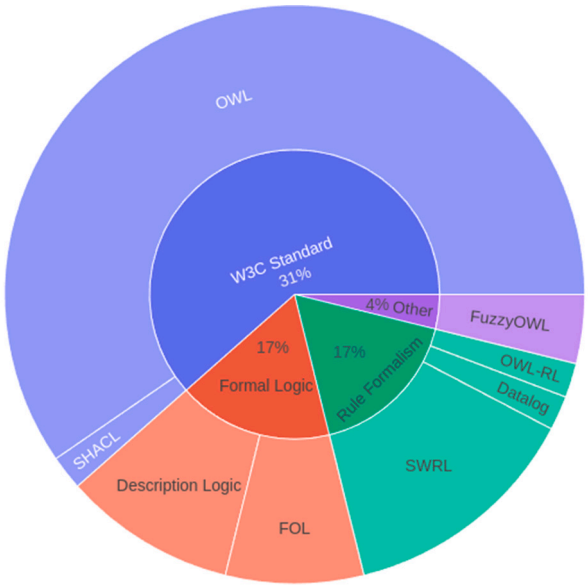


Fig. 11. Number of papers targeting each formal language to describe constrains as axioms and rules within an ontology (Answer to RQ2.1).

class axioms [29,73,74,77], class expression subsumption [26,79] and complex rules [75,76,78]. **Prescriptive document** (in green) is used specifically to identify rules [28,30,82], apart from one article [31], learning object property axioms.

This section has reviewed the different axioms and rules available in the literature, answering RQ2.1, and proposed a refined classification of the Ontology Learning Layer Cake for the last two layers (axioms schemata and general axioms). The distribution of the axioms across the surveyed paper has been analyzed by identifying the most (Subsumption, cardinality, disjointness) and least (role inclusion, property chain, asymmetry) targeted axioms. It also proposed a comparative analysis of the type of axioms learned from each format and content of the data (as defined in Sections 4.1.1 and 4.1.2). The sections below will address the language chosen by the literature to express the axioms and rules (Section 4.2.2), the level of expressiveness (Section 4.2.3), and the objectives of the paper to learn those axioms (Section 4.2.4).

4.2.2. Targeted formal language(RQ2.2)

This section presents the different formal languages covered in the literature to integrate axioms and rules within an ontology. The main formal language used in the literature is the OWL and OWL2 standard as defined by the W3C standard, which is also the analytical framework of this section (see Section 4.2.1), however, some papers of this survey chose other formal languages to express the constraints over the ontology, either with another rule standard (SWRL, OWL-RL, etc.), other deductive languages (Datalog) or logic syntax (Description Logic, First-Order-Logic).

Fig. 11 presents the distribution of the different languages targeted by the paper within the literature.

OWL and OWL2, the Web Ontology Language defined by the W3C,[19] is the most used formal language, within more than half (58%) of the surveyed papers [26,29,36–45,47,51,54–56,61–63,65–68,72–74,77,79,80,83] **SWRL and OWL-RL** are two propositions within the W3C consortium to integrate complex rules within the ontology framework. SWRL[20] is "based on a combination of the OWL DL and OWL Lite sublanguages of the OWL Web Ontology Language with the Unary/Binary Datalog RuleML sublanguages of the Rule Markup Language". It is targeted in seven papers [28,30,52,53,70,75,78]. OWL-RL[21] is an OWL2 profile that can be expressed using rule-based technology, as defined by the RIF format. It is targeted in one paper [60]. **SHACL**, the Shapes Constraint Language defined by the W3C[22] is used by one article [50]. **Datalog** is a declarative logic programming language, a subset of Prolog which is used to perform declarative queries over a database. It is targeted in one paper [34] **Formal Logic** languages like Description Logic [31,35,46,58,59] and First-Order Logic [48,57,71,76] are also targeted in the literature, without any precision on their use for implementing an ontology. **Fuzzy OWL** is a proposed formalism to integrate Fuzzy Logic within the ontology format. It is targeted in [49,69].

Table 5 provides an overview of the different formal languages found in this analysis and their corresponding articles.

Table 5
Formal Languages and the corresponding articles referring to them (Answer to RQ2.2).

Language-family	Language	Articles
W3C Standard	OWL	[26,29,36–45,47,51,54–56,61–63,65–68,72–74,77,79,80,83]
	SHACL	[28,30,52,53,70,75,78]
Rule Formalism	SWRL	[60]
	OWL-RL	[50]
	Datalog	[34]
Formal Logic	FOL	[49,69]
	Description Logic	[31,35,46,58,59]
Other	FuzzyOWL	[48,57,71,76]

4.2.3. Level of expressiveness (RQ2.3)

This section presents the level of expressiveness that is explicitly targeted by the reviewed papers. The level of expressiveness of a description logic defines the type of inferences that can be expressed by the ontological system. It is a trade-off with the decidability of the system, as more expressive ontology will lead to undecidable statements. Within the surveyed literature, only 15 articles express explicitly the level of expressiveness addressed, which corresponds to a bit less than 30% of the reviewed papers. This proportion of papers explicitly stating the level of expressiveness is higher than other research papers in the

[19] https://www.w3.org/TR/owl2-syntax/
[20] https://www.w3.org/submissions/SWRL/
[21] https://www.w3.org/TR/owl2-profiles/#OWL_2_RL
[22] https://www.w3.org/TR/shacl/

field or other research in the combination of Machine Learning and Ontology (as stated by the review done by Ghidalia et al. [85]). This is mainly due to the choice of our subset, which targets more precisely a community aware of the different axioms of Description Logic.

The targeted level of expressiveness ranges from $\mathcal{ALC}$ to $\mathcal{SHOIN}(\mathcal{D})$ available for OWL-DL. The different levels of expressiveness found in the surveyed literature are the following:

$\mathcal{ALC}$ [54,59,79] is a light description logic that integrates the union of classes, complements class to the $\mathcal{AL}$ level with the intersection of classes, class inclusion, and equivalence. $\mathcal{EL}\bot$ [58] is another light description logic that includes the union of classes, existential restriction, and the bottom operator. $\mathcal{ELO}$ [46] is another variation of the $\mathcal{EL}$ description logic, including nominals within a list. $\mathcal{ALCI}$ [62] integrates inverse properties to the previous $\mathcal{AL}$ level defined previously. $\mathcal{ALCQ}$ [26,77] adds qualified and unqualified cardinality restriction to the $\mathcal{ALC}$ profile. $\mathcal{ALCO}(\mathcal{D})$ [43] includes named individuals within a list and datatypes within the $\mathcal{ALC}$. $\mathcal{SROIQ}$ [60] includes concept inclusion and intersection, transitivity, role inclusion, reflexivity and symmetry, disjoint roles, nominals, inverse, and cardinality restriction. $\mathcal{SHROIQ}$ [74] includes concept inclusion and intersection, transitivity, role inclusion, reflexivity and symmetry, disjoint roles, nominals, inverse and cardinality restriction.

There is a special mention for [52], which targets DL-Safe, a restriction over the SWRL rule language that is constructed to guarantee the decidability of the rules by restricting the rules to individuals explicitly expressed within the knowledge base.

4.2.4. Objective of learning expressive axioms (RQ2.4)

The studied papers pursue different objectives to learn expressive axioms and rules. The objectives expressed by the authors are the following:

Create a new ontology from scratch: create the ontological representation from the available sources of data describing the domain without any existing ontological resource.

The goal is to capture the description of the domain and create a first ontological representation using available resources in any format [26,29,61,63–67,69,75,77–80,82].

Complete the knowledge Base: complete an existing knowledge base with more expressive axioms. This objective is put forward by papers learning from encyclopedic knowledge bases such as DBpedia another generic knowledge base, with the explicit aim to improve the quality of the encyclopedic knowledge or domain ontology for further use [28,30,31,36,37,39–41,43–45,47,49–51,54–56,58–60,73,74,81].

Find rules and classification in data: identify statistically significant relations between examples

in order to apprehend unforeseen information to complete the expert knowledge [34,35,38,48,52,53,70–72].

Evaluate axioms and rules: evaluate the relevance of candidate axioms and rules before integrating them into the knowledge base or to prune the knowledge base from unnecessary or redundant information to propose more relevant inference [42,46,57].

Make inferences: create an ontology to propose relevant inferences, recommendations, or predictions [28,68,76].

Map ontology to source data and validate the consistency of the mapping: in the context of creating an ontology from a relational database, keeping track of the relation between the source database and the ontology is mandatory to assure reliable maintenance over time of both systems [62,66].

Table 6 provides a summary of the objectives presented in the reviewed literature and the papers associated to each.

In addressing RQ2(What are the literature articles learning?), we seek to study the predominant expressive axioms and rules learned from the literature and the source data used to learn these axioms. To comprehensively address this question, RQ2.1 (Section 4.2.1) delved into the types of axioms and rules targeted by literature articles and classified them in Table 3 mapping ontology learning layers to OWL2 corresponding axioms. We noticed that Object Property and Class Expression are the two main axioms learned from the literature, while only a few articles learn rules (see Fig. 8). Learning these two axioms relies heavily on Knowledge Bases (see Fig. 9) and example data (see Fig. 10). Axioms are usually expressed using a formal language. RQ2.2 (Section 4.2.2) provided an overview of the various formal languages employed for axiom learning. While the literature mainly follows W3C stands (62% of the papers), other formal languages such as SWRL, OWL-RL, and Datalog are also utilized. Additionally, SHACL, Datalog, Description Logic, First-Order Logic, and Fuzzy OWL are among the targeted formal languages. Moreover, RQ2.3 (Section 4.2.3) presented the explicit expressiveness levels targeted by the reviewed papers. We only found 15 articles explicitly stating the targeted expressiveness level, about 30% of the articles. We found in the literature an expressiveness level ranging from $\mathcal{ALC}$ to $\mathcal{SHOIN}(\mathcal{D})$ for OWL-DL, with variations like $\mathcal{ALC}$, $\mathcal{EL}\bot$, $\mathcal{ELO}$. Finally, RQ2.4 (Section 4.2.4) outlined various objectives pursued by research papers in learning expressive axioms and rules. These objectives included creating a new ontology, completing a knowledge base, evaluating the relevance of axioms in an ontology, and improving inference mechanisms, or mapping ontologies between them.

Table 6
Objective of the articles (Answer to RQ2.4).

Objectives	Articles
Axiom and rule evaluation	[42,46,57]
Create new ontology from scratch	[26,29,61,63–65,67,69,75,77–80,82]
Find rules and classification from data	[34,35,38,48,52,53,70–72]
Knowledge Base completion	[30,31,36,37,39–41,43–45,47,49–51,54–56,58–60,73,74,81]
Make inferences	[28,68,76]
Map ontology to source data	[62,66]

4.3. How do they learn? (RQ3)

Different algorithms from the field of Artificial Intelligence are used to process diverse data sources and achieve targeted expressiveness, usually classified as rule-based, unsupervised learning and (semi-)supervised learning algorithms. This section aims to answer the question RQ3: *How are they learning?*, which focuses on the process used to extract axioms from data. This section divides the analysis into four subquestions: **RQ3.1**: what kinds of algorithms are used?; **RQ3.2**: how are the algorithm evaluated?; **RQ3.3**: What technical framework are used to implement the solution?; **RQ3.4**: Is there an available implementation for reusing the solution?

Section 4.3.1 reviews the literature concerning the type of algorithms used. Section 4.3.3 analyzes the development frameworks in terms of programming languages (Java, Python, C#), tools, and libraries used by the authors. Section 4.3.2 surveys the type of evaluation undergone within the literature to measure the efficiency of their solution in terms of evaluation techniques. The evaluation metrics are also examined to understand the statistical tools used to confirm the validity of the solution. Finally, Section 4.3.4 reviews the available tools and implementations identified in the literature.

4.3.1. Ontology learning algorithms (RQ3.1)

Different algorithms are used in the literature to tackle the problem of expressive ontology learning. These algorithms are divided into four families: Rule-based algorithms, Unsupervised Learning Algorithms, Evolutionary Algorithms, and Neural Network Algorithms.

Rule-based algorithms are based on predefined rules to analyze the information of the data. Those rules are specific to each kind of data and task. The main used rules are: mapping rules [61,63,65–69]

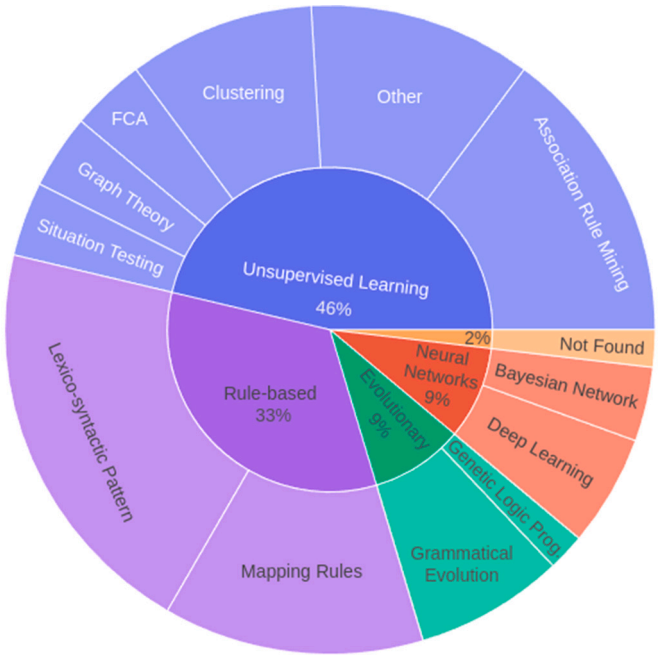


Fig. 12. The distribution of the articles by algorithm type (Answer to RQ3.1).

for correspondence relations between database and knowledge base, lexico-syntactic pattern [28–31,73,74,76–79,81] to map the element of the natural language to the knowledge base.

Unsupervised Learning Algorithms are grounded on different kinds of mathematical and statistical methods to determine the appropriate group of examples that would have a statistical significance to determine a rule or an axiom. This is the most diverse group of algorithms with Association Rule Mining [51,52,55,57,60,70–72], Graph Theory [48,62], Clustering [38,39,42,49,56,83], Situation testing [46,53], Formal Concept Analysis [35,58].

Evolutionary Algorithm are inspired by biological evolution and iterate over several generations to build candidate rules and axioms that are selected by a fitness function. This family includes two types of algorithms, Grammatical Evolution [37,41,44,45,47] and Genetic Logic Programming [34].

Neural Network Algorithms are based on an architecture of neurons, as inspired by the neural cells in the brain, which correspond to nodes in a graph layer, where nodes and edges are weighted, those weights being adjusted during the learning phase. Neural Network architectures have exploded [85] during the past years and come into many forms. We identified two specific types in our review: Bayesian Network [54,59] and Deep Learning [26,40]. Bayesian Networks are neural networks adjusted based on the Bayesian formula of conditional probability. Deep Learning is the most recent architecture of a Neural Network, with several layers of neurons connected together.

Fig. 12 presents the distribution of algorithms used in the literature, which is mostly divided between Rule-based and Unsupervised Learning algorithms. Together they cover over 3/4 of all the articles (around 33% for rule-based algorithms and 46% for unsupervised algorithms). It is quite surprising that Neural Network algorithms have not been used a lot to tackle the issue at hand, even though it has been a hot topic in Computer Science for the past years.

Moreover, Fig. 13 presents the evolution of the algorithms over time. The rise of evolutionary algorithms is quite visible in 2019, as led by the work of A. Tettamanzi and his team [37,41,44,45,47,52,83], which seems to have reached a new state of the art for many different axioms (as we will discuss underneath). However, they have not replaced other algorithms but have mostly enriched the variety of algorithms that can be used to identify axioms.

When we examine more closely in which case each kind of algorithm is used, as presented in Fig. 14, there is a quite neat distinction between rule-based algorithms and unsupervised neural networks and evolutionary algorithms based on the preferred kind of information they will target:

Descriptive and prescriptive documents are mostly treated with rule-based algorithms, being mapping-rules for database [61,63,65–68] or lexico-syntactic patterns for textual documents [29,73,74,76,77,79]. One noticeable exception is [26], which explicitly framed the task of learning expressive ontology as a Translation task as a standard task in the NLP community. **Example data** on the other hand is mostly addressed from unsupervised learning techniques [36,38,39,43,46,48–53,55–58,70–72,80,83], from neural networks [40,54,54,59,59] and more recently from evolutionary algorithms [34,37,41,44,45,47].

The distribution of the different kinds of algorithms is marked more strongly from the type of information than from the type of data in the source, as presented in Fig. 15. This is due to the diversity of information identified in each type of data: when the algorithms analyze examples presented in text or database structure, the kind of analysis is very similar to what is required, for instance, in a knowledge base.

The summary of the different algorithms can be found in Table 7.

It is quite remarkable to note that the type of axioms targeted, on the contrary, does not define much of the specific family of algorithms used, as shown in Fig. 16. The distribution of axioms is quite well arranged over the different kinds of algorithms. However, we can notice some tendencies:

Rule-based algorithms are mostly used to learn object property axioms (mostly from databases schema) [31,61,65–67], and the other half is interested in learning class axiom [29,73,74,77], class expression [63,69,79], and complex rules [28,30,76]. **Unsupervised learning** is mainly used to learn complex rules [48,52,53,70–72,75,82], as well as class expression [38,39,43,46,58,80,83] and object property axioms [36,51,57,62]. **Evolutionary algorithms** are mainly used to learn class axioms [41,44,45,47], but they were also tested to learn class expression ([37] and complex rules [34]. **Neural networks** were mainly explored to learn class expression [26,54] and tested for class axiom [59] and object property axioms [40].

4.3.2. Evaluation of algorithm (RQ3.2)

The evaluation of the ontology learning process is a crucial task to define the relevance of the learned information to create or improve an ontology [11]. Different types of evaluation were identified in [11] and are also used within the surveyed papers:

Task-based evaluation: The accuracy of the created ontology is addressed based on a specific task, the ontology is created to make a recommendation or infer some knowledge [31,46,73,76].

Gold Standard: The learned axioms are compared to an existing standard created by a human expert, which is used as a reference to compare the performance of the algorithm [40,45,47,53,57,61,63,66,75,76,80].

Benchmark: Similar to the gold Standard, but the algorithms are evaluated on a set of different gold standard data, and a comparison with different algorithms is performed [26,29,34,36,38,41,43,44,46,48,51,52,54,55,59,74,77]. This is the highest level of evaluation, to be able to define the state of the art of a given task.

Criteria-based evaluation: The learned axioms are evaluated based on defined criteria associated with the initial data, for example, a correspondence to some syntactic structure, coherence evaluation, or statistical relevance [37,58,69,70,72]. These measures are used either within the training or as final evaluation, when there is no gold standard to define the exact solution.

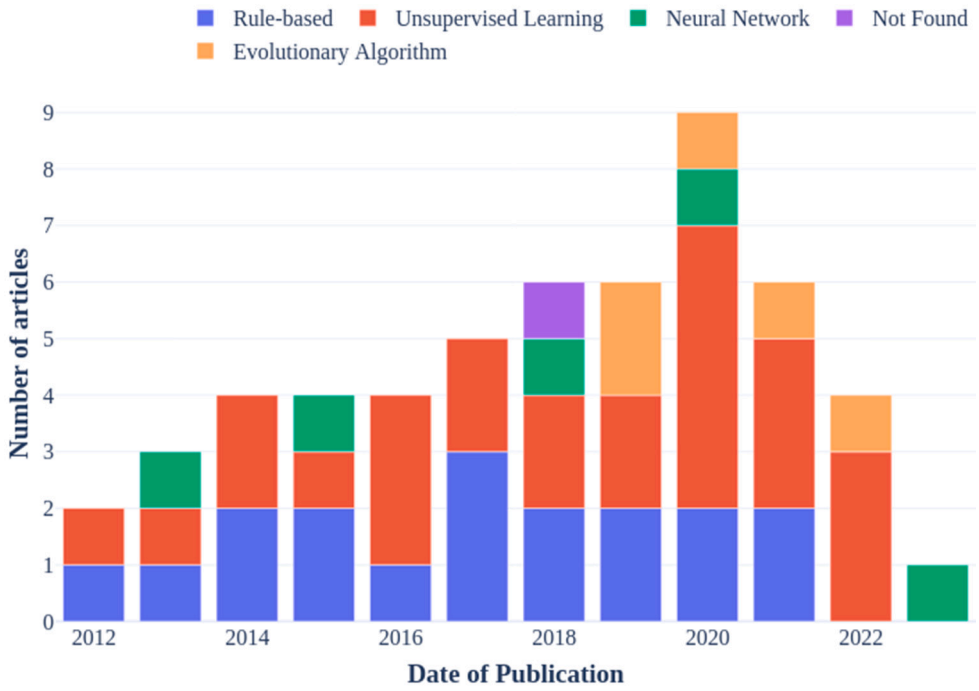


Fig. 13. The evolution of algorithms type over time (Answer to RQ3.1).

Table 7
Articles depending on the algorithms used for the different data format and content (Answer to RQ3.1).

Content type	DataType	Algorithm	Articles
Descriptive data	RDB	NF	[64]
		Rule-based	[61,63,65–69]
		Unsupervised Learning	[62]
	Text	Neural Network	[26]
		Rule-based	[29,73,74,76–79]
		Unsupervised Learning	[75]
Example data	Knowledge Base	Evolutionary Algorithm	[34,37,41,44,45,47]
		Neural Network	[40,54,59]
		Unsupervised Learning	[35,36,38,39,42,43,46,48–53,55–58,60]
	RDB	Unsupervised Learning	[70–72]
	Text	Rule-based	[81]
		Unsupervised Learning	[80]
Prescriptive data	Text	Rule-based	[28,30,31]
		Unsupervised Learning	[82]

Manual evaluation: When the condition did not provide a satisfactory source of truth to undergo a complete evaluation, authors evaluate the quality of the created ontology with their own assessment [30,60,62,81].

Several criteria to determine the relevance of the constructed ontology compared to a target or an expected standard have been proposed in the literature. Our analysis distinguishes between several kinds of criteria, depending on their purpose:

Firstly, **criteria to evaluate the accuracy of the executed task** compared to a proposed standard correspond to usual metrics in supervised Machine Learning literature: precision [29,30,34,36,38,41,45,46,48,51,52,54,55,59,61,63,66,73,74,76], recall [30,38,45–47,51,54,55,59,61,63,66,73,74], accuracy [26,41,47,54,55,60,72,81], F-measure [30,38,54,55,63,73] and F1-score [59,61,74,80], AUC [40,54], fitness [37], RMSE [83], standard deviation [83], ROC [54], error-rate [54]. **Criteria to evaluate the relevance of the measure** are proposed when there is no standard, by assessing the internal quality of the proposed rule or axiom in terms of statistical support: Confidence [34,36,52,53,57,58,70,71], Support [36,50,52,53,57,70,71], coverage [31,34,47,52,75,76], PCA [34,71], chi-square [70], frequency [39], consistency, credibility. Other **criteria are defined to evaluate the accuracy of the text**: as textual data is quite specific because the same meaning can be expressed with different words and grammatical structures, several metrics have been proposed in the NLP community to assess the quality of the task: BLEU [31], grammatical coverage [77], similarity [44], edit distance [26], correctness [77], generality and credibility [44]. **Criteria are defined to evaluate the logical structure** within the created knowledge base: consistency [62,72], Mingen Measure [57], Partial implication [58] Finally, **technical evaluation** assesses the performance of the algorithm in terms of the resources they require on the machines to run: runtime [34,35,43,48,50,69], memory footprint [34].

Table 8 displays the different metrics used in the literature.

4.3.3. Technical framework and language of development (RQ3.3)

Fig. 17 displays the main programming languages used to implement the proposed solution: Java, Python, C#. Different frameworks and tools are also used within the literature.

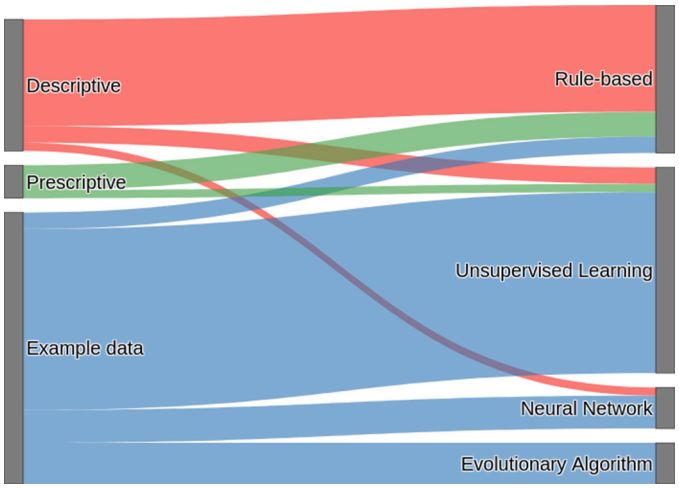


Fig. 14. The distribution of the articles by algorithm type depending on the type of information they analyze (Answer to RQ3.1).

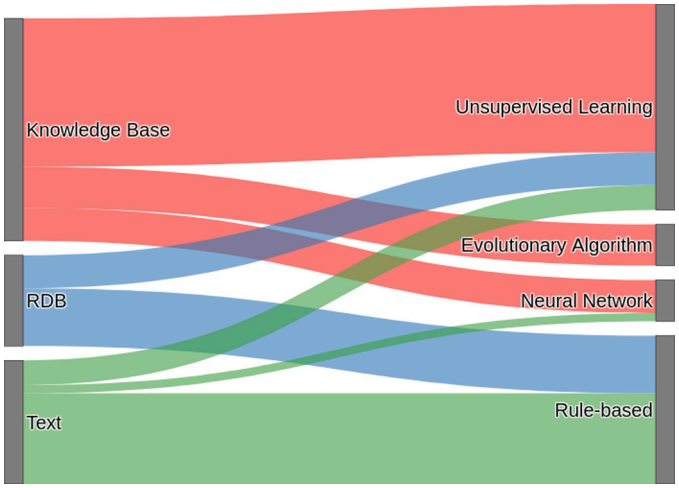


Fig. 15. The distribution of the articles by algorithm type depending on the source format (Answer to RQ3.1).

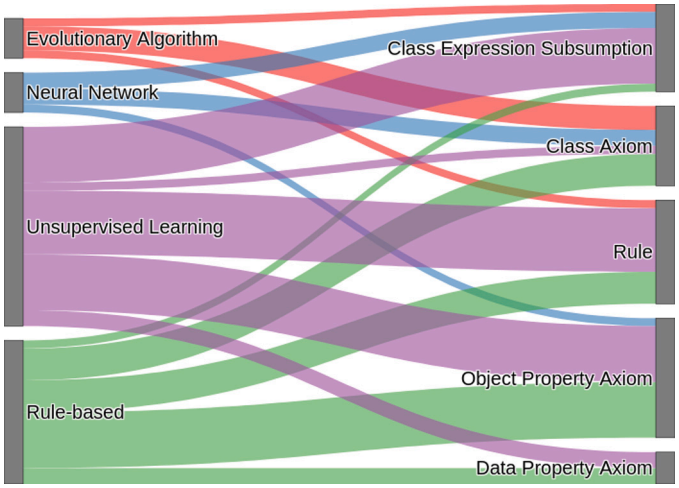


Fig. 16. The distribution of the articles by algorithm type depending on the type of axioms they learn (Answer to RQ3.1).

Table 8
Evaluation metrics used in the literature papers (Answer to RQ3.2).

Metric type	Metric	Articles
Supervised metrics	Precision	[29,30,34,36,38,41,45,46,48,51,52,54,55,59,61,63,66,73,74,76]
	Recall	[30,38,45–47,51,54,55,59,61,63,66,73,74]
	Accuracy	[26,41,47,54,55,60,72,81]
	F-measure	[30,38,54,55,63,73]
	F1-score	[59,61,74,80]
	AUC	[40,54]
	Fitness	[37]
	Error-rate	[54]
	Average match rate	[43]
	ROC	[54]
	RMSE	[83]
	Standard deviation	[83]
Statistical metrics	Confidence	[34,36,52,53,57,58,70,71]
	Support	[36,50,52,53,57,70,71]
	Coverage	[31,34,47,52,75,76]
	PCA	[34,71]
	Frequency	[39]
	Kernel Density estimation	[39]
	chi-square	[70]
Logical metrics	Consistency	[62,72]
	Mingen measure	[57]
	Partial implication (LUX)	[58]
Textual metrics	Credibility	[44]
	Generality	[44]
	Similarity	[44]
	Correctness	[77]
	Grammatical coverage	[77]
	Edit distance	[26]
Task-oriented	Median	[46,83]
	Average	[46]
Technical metrics	Runtime	[34,35,43,48,50,69]
	Memory footprint	[34]

Several **Ontology and Knowledge Graph Editors** have been created to help manually develop ontologies. Those tools are referred to in the literature to create a first ontology which will guide the ontology learning process or to display the final ontology: Protégé [30,50,53,55,63,70,72,75], Weka Toolkit [55], OntoGraph [30], Neo4j [70]. **Rules engines** are used within those tools to run the rules and infer the appropriate deductions over the knowledge base: HermiT [72], Jena [50]. The natural language community have created over the years several libraries and tools (**Natural Language Framework and Parsers**) to reuse existing algorithms in further work easily: Stanford NLP [28,31,73,74,76,79], Watson NLU [73], ACE Parser [77], WordNet [78,79], Malt Parser [76], Metamap [80], Spacy [30], Gensim [28], OpenCalais [29,73]. **Machine Learning libraries** correspond to the same objective to create on-the-shelves tools to facilitate training of machine learning algorithms: TensorFlow [26,40], Java-ML [49]. **Technical support tools** simplify the execution of the algorithm, especially via cloud architecture: Virtuoso [37,41,44,48,60], Google Collaboratory [70], Google Web Toolkit [56], DDLutils [67]. Other algorithms for Ontology Learning are reused within our surveyed literature to execute part of the pipeline of the learning algorithm: AMIE+ [51], GoldMiner [55], Swift Link Data Miner [39].

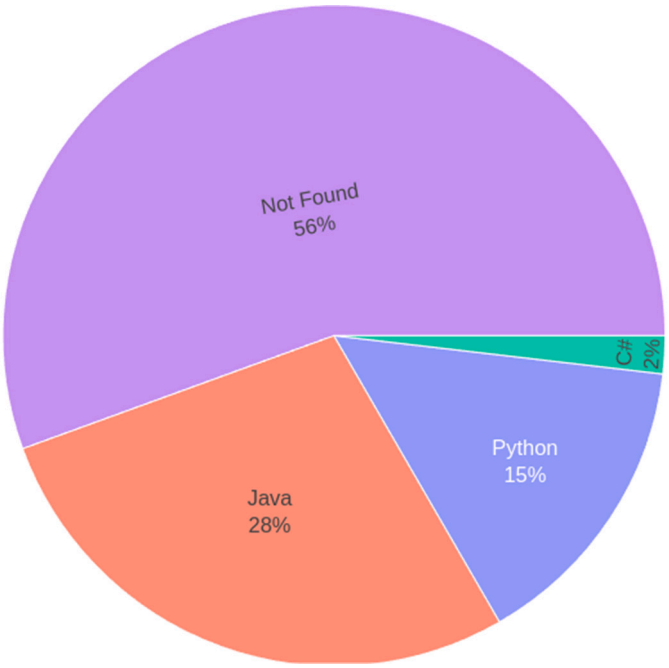


Fig. 17. The distribution of the articles by programming language used (Answer to RQ3.3).

Table 9
Specific algorithms created to address each data format (Answer to RQ3.4).

Data type	Algorithms
RDB	W-GRaph [62], DBToXML DBToOWL [67], CPAR [72], C2Ont [63], OWLRel [66], AMIE [71]
Knowledge base	LeDA [55], RuleGenerator [53], Datil [49], TAMO [46], GoldMiner [60], BelNet [59], ProLOD++ [56], RuDik [48], DL-FOIL/DL-FOCL [43], Evoda [34], Swift Linked Data Miner [50], BelNet+ [54], Multi-Objective Evolutionary Algorithm [44]
Text	source2onto [75], TEDEI Text Description Identifier [77], NALDO [74]

4.3.4. Available tools and algorithms (RQ3.4)

This section presents the tools and algorithms available in the surveyed literature when they are presented by the authors in a reusable fashion, through a tool or as a named algorithm.

A few of the reviewed papers provide a direct implementation of their algorithms in existing tools and frameworks, which can, therefore, be reused easily. The 3 main tools for ontology learning are the following:

Protégé plugin: Embed to the ontology editor Protégé to integrate the work directly in one of the most used editors; **DL-Learner**[23] is an open-source Java tool proposed by [86] which can be associated to the main ontology editors; **OntoLearn**[24] Python library developed by the Dice group at Paderborn University.

Part of the surveyed literature presented their work with a named algorithm to help reference their work in the future. Those algorithms are most of the time specific to the source data they address and, therefore, are classified as such.

Table 9 presents the current algorithms available in the literature and the types of data they address.

This section answered the RQ3 (How are they learning?), showing the families of machine learning algorithms, the evaluation strategies, the programming languages, and the available tools for ontology learning. Q3.1 (Section 4.3.1) explored the algorithms used for ontology learning, namely Rule-based and Unsupervised Learning, Association Rule Mining, Graph Theory, Grammatical Evolution, Genetic Logic Programming, and Neural Networks. The distribution of these algorithms shows Rule-based and Unsupervised Learning as the predominant ones. Surprisingly, Neural Network algorithms have limited use in expressive ontology learning. We noticed that rule-based algorithms are predominantly used for descriptive and prescriptive documents, whereas unsupervised learning techniques are favored for example data (see Fig. 14). We also found the predominant use of Rule-based algorithms is for learning Object Property and Class axioms. In contrast, Unsupervised algorithms are mainly used to learn Class Expression Object Property axioms (see Fig. 16). To evaluate the quality of the algorithms, Q3.2 (Section 4.3.2) outlined the main evaluation methods, including task-based evaluation, gold standard comparison, benchmarking, criteria-based evaluation, and manual assessment. These approaches encompass assessing ontology accuracy, comparing learned axioms with existing standards, evaluating against benchmark datasets, and considering criteria such as syntactic structure correspondence and statistical relevance. Furthermore, diverse evaluation criteria, such as spanning precision, recall, accuracy, and text-specific metrics, alongside technical assessments like runtime and memory footprint, are employed to gauge ontology learning efficacy. Moreover, Q3.3 (Section 4.3.3) analyzed the development frameworks and libraries used in the reviewed papers. Protégé, Weka Toolkit, OntoGraph, and Neo4j are used for ontologies management. Tools such as HermiT and Jena enable the execution of rules and inference. Only 57% of the articles provided the programming language, and 40% of the paper relied on Java, while only 15% used Python. Meanwhile, software frameworks such as Spacy, Gensim, OpenCalais, TensorFlow, and Java-ML are used for natural language processing and training machine learning algorithms. Finally, Q3.3 (Section 4.3.4) provided an overview of tools available in the literature for ontology learning, namely the Protégé plugin, DL-Learner, and OntoLearn. The majority of the tools are implemented for articles learning from Knowledge Bases.

5. Discussion and research directions

5.1. Discussion

We found that the status of the research is quite different according to the different data and content formats :

Learning from the database description seems to be at the maturity stage with mapping rules. Most of the works seem to have been done during the 2000s (outside the temporal scope of this paper), and the few remaining articles address questions like adding flexibility to the process. However, we have not found any standard tool or library that tackles this topic. **Learning from the example in the database** is based on induction algorithms similar to the ones learning from the knowledge base. Theoretically, learning from database examples can essentially be distilled into a problem of learning from the knowledge base once the connections between the database schema elements and the ontology have been established. **Learning from knowledge base examples** is the most mature, with several algorithms (GoldMiner [60], BelNet & BelNet+ [54,59], LeDA [55], RuDik [48], Evoda [34], AMIE+ [51,87]) and several existing tools and library (OntoLearn, DL-Learner, plugging Protégé). Most of the articles are learning axioms from the DBPedia dataset, using its existing structure as a Gold standard. The algorithms are mature enough to be compared with benchmark datasets. **Learning from the description in the text** seems to be still at the investigation level. A state of the art was reached around 2015 with lexico-syntactic patterns to use the sentence's grammatical structure to identify elements of the ontology.

A new research direction was defined with the recent advances in Neural Network technologies for NLP, which consider ontology

[23] http://dl-learner.org/
[24] https://github.com/dice-group/Ontolearn

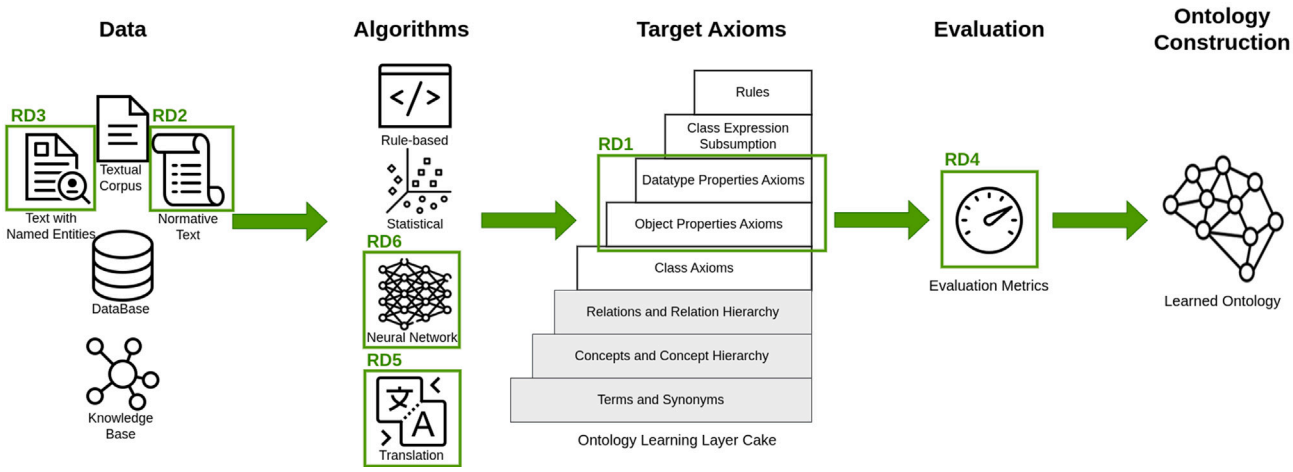


Fig. 18. 6 Research Directions are identified within the Ontology Learning Process: *RD1*- Target ObjectProperty and DataType Properties axioms, *RD2*- Address prescriptive documents, *RD3*- Identify named instances in text, *RD4*- Improve evaluation of axioms and rules, *RD5*- Translate complex rules from text, *RD6*- Explore the capacity of neural networks. Those directions may be explored one-by-one or combined together to improve the capacity of the process.

learning a translation task from Natural Language to Logic, as proposed by [26]. **Learning from examples in text** is based on identifying Named Entity in the text and extracting their relations, using Named Entity Recognition and lexico-syntactic patterns. The identified entities are then injected within a knowledge base, and the axioms are learned with inductive techniques from the examples in the knowledge base.

Despite the overall quality of the articles reviewed, which explicit description of their motivation, their positioning, the model they used, and the solution they proposed, we have to underline two foremost critics of the quality of the overall corpus: the code and data are seldom open source, which undermines the process of replication and the level of expressiveness of the expected ontology is not always stated, nor the motivation to target such level.

Concerning the level of expressiveness of Description Logic, only 14 articles of 52 explicitly state the level they target as presented in Section 4.2.3. As an upside remark, we can highlight that this specific subset of ontology learning is much more explicit towards their level of expressiveness than other research papers in the field or other research in the combination of Machine Learning and Ontology (as stated by the review done by Ghidalia et al. [85]). This is mainly due to the choice of our subset, which targets a community that is more precisely aware of the different axioms of Description Logic. However, we would like to emphasize the absence of motivation for some work as the choice of each level of Description Logic is not explicitly stated nor explained. In particular, the level of Description Logic is always a trade-off between expressiveness and decidability, and most of the articles take for granted that going for more explicit axioms will necessarily improve the quality of the ontology, with little mention of the targeted objective behind the construction or improvement of the ontology. If this lack of motivation is understandable for work on a general knowledge base like DBPedia, where the expressiveness could be maximal to express all the knowledge gathered across all domains, this is more problematic for work addressing domain ontology to make inferences over the data.

There seem to be two standards in the literature concerning open data and open source. Papers learning from examples in the knowledge base often use existing open-source datasets, such as DBPedia. However, sometimes, finding the exact version of DBpedia is difficult. There are also existing tools and algorithms at the state-of-the-art level (DL-Learner, OntoLearn, and Protégé Plugin). Those tools are open-source and often updated with new algorithms. On the other hand, it is a real loss for the community that no other tools are currently maintained and available, especially for learning axioms and rules from text or databases. The same is true for datasets, which are not often shared and are quite often specific to a given task.

A more general criticism of the articles concerns the evaluation of the created ontology. Very few papers evaluate their work based on a specific task and objective. A good example of a task-based evaluation is the paper on Avionics Maintenance [46], where the learned rules are selected based on the most specific (and lowest number) of recommendations it infers. Very few papers are as evident in their objective of learning and evaluating each axiom. Moreover, there is a real challenge around the evaluation at different levels: external and internal. For external evaluation, evaluating the accuracy of the task is a real challenge without a proper benchmark dataset. As for internal evaluation, assessing the coherence and the appropriate inclusion of an axiom within the Knowledge Base is a challenge some authors address, especially when selecting the appropriate rules and axioms. Furthermore, there is almost no discussion over the quality of the learned representation: it is assumed that the choice of predicate (Class, Object Property, or Data Property) is already given or is not an issue for the task. It is especially problematic for the articles learning rules from textual documents, as they are most of the time grounding their choice on the grammatical structure of the sentence without further consideration of the potential ambiguity of this grammar (should a name always be a Class and a verb a Relation ?)

5.2. Opened research directions

At the end of this analysis, we identify several open research directions for learning expressive ontology. Fig. 18 illustrates the process of ontology learning and highlights the different research directions identified during our analysis. All studies goes through the four steps of the Ontology Learning Process(data, algorithms, target axioms and evaluation) as described within our survey. The research directions emphasize the parts of the process which are worth exploring one-by-one or together to improve the overall quality of the process. Those directions are either mentioned in the reviewed papers or identified based on comparison with other Machine Learning fields:

RD1 Identify axioms not yet addressed in the articles, like asymmetry, min–max values, property chain, role inclusion, and datatypes. Unsupervised algorithms like clustering algorithms may be a good lead for investigating this direction. Genetic Programming is also on the rise to address all axioms from the W3C classification.

RD2 Target prescriptive documents such as regulatory documents as a specific source of information that provides ground truth rules of a domain in a semi-structured manner. Those documents, by

their format, integrate many rules and have a well-defined structure that can facilitate the ontology learning process. Therefore, it is a qualitative data source that simplifies learning expressive ontology by the quality of the data and the precision of the terms employed.

RD3 Identify instances in the text to learn class and object properties axioms: Named-Entity Recognition (NER) provides some great techniques to analyze a vast corpus of data available in textual format, in documents or web pages, and transform them into instances of a knowledge base. From these instances, any unsupervised learning algorithms can be applied to identify new axioms, such as class axioms or object properties axioms. Therefore, this process can increase example data for learning expressive ontology.

RD4 Evaluation of axioms and rules learned: the ontology description is always a trade-off between expressiveness and decidability, and there may be some expressions that are semantically equivalent on our knowledge base but may provide too much or too little information to complete the task at hand. Therefore, selecting the appropriate axioms and rules to tackle the task properly is still an open research question that is mandatory to improve the quality of the ontology.

RD5 Translate complex rules from textual description into Logical format to integrate within the ontology: the semantic of the component of the natural language description, such as verb and noun, could be interpreted to identify that two sentences mentioning "produce" or "make" convey the same relational meaning and that the former should be preferred to the later as it is more specific. Some work is already engaged on this topic, but it often relies on specific syntactic structures that must be described manually and may vary between languages. One open search is to incorporate the meaning of the words within the translation of rules into formal logic to complete the analysis of the syntactic structure [26].

RD6 Explore the capacity of Neural Networks to learn rules and axioms. More specifically, the recent progress in NLP with the introduction of the Transformers architecture and LLM seems to be a promising direction to address the challenge of analyzing natural language resources. Therefore, this direction involves exploring using LLM to extract first-order logic formulas from textual documents.

6. Conclusions

Ontologies and associated rules enable logical inferences and decision-making processes in machines, making interactions with complex data systems more intuitive and effective. They also bridge human knowledge and machine understanding, allowing more accurate and automated information retrieval and decision support systems. This capability enhances machine understanding and reasoning in diverse applications, from artificial intelligence to data integration. Our paper's research on ontology learning highlights the challenges and significant advancements in the field of conceptual representation for machines. Our analytical work has thoroughly examined the various methods and tools used to extract and formalize knowledge from heterogeneous information sources, including textual documents and databases. Inspired by the systematic literature review methodology, our study has summarized the existing approaches and critically evaluated their effectiveness and limitations. Our review has underscored the potential of ontology learning, particularly at the top level of the Ontology Learning Layer Cake: rule identification. This identification facilitates the derivation of general hypotheses and complex relationships, which is essential for creating more expressive and semantically rich ontologies.

Beyond the scope of our study, we recommend delving into the detailed research by Ozaki [88] to explore the theoretical aspects underpinning ontology learning. They formalize five approaches (Association Rule Mining, Formal Concept Analysis, Inductive Logic Programming, learning theory, and neural networks) for deriving description logic axioms from example data within a knowledge base. While this theoretical exploration lays a solid mathematical foundation for understanding the challenges involved, it stops short of evaluating these approaches' practical application and outcomes, as we provided in our study.

Our study identified several promising research directions. These include exploring unsupervised learning techniques for axiom identification, applying machine learning to regulatory texts to extract domain-specific rules, and harnessing advancements in natural language processing to translate complex textual information into logical formats suitable for ontological frameworks. Additionally, the potential of neural networks, especially recent developments in transformer architectures, offers exciting prospects for extracting and formalizing logical rules directly from textual data. The field of ontology learning is crossing the boundary from research into the industrial world of software engineering [89]. To ensure the broader adoption of these research methods by industrial groups, the automation, and scalability of ontology learning methods must be combined with noise handling and precision improvement [89,90]. This integration will enable industries to reconstruct the know-how in databases or electronic document management systems more effectively [91].

This study contributes to our comprehension of the ontology learning process and highlights the critical need for continued innovation. Developing more automated, accurate, and contextually aware systems for knowledge representation becomes essential for the next generation of intelligent systems, ensuring they can operate effectively within human knowledge's complex and nuanced realms.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Armary reports financial support was provided by Anabasis Assets. Armary reports financial support was provided by French National Research and Technology Agency. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly and Microsoft Copilot in order to correct grammar and spelling mistakes. After using those tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Data availability

Data will be made available on request.

References

[1] T.R. Gruber, A translation approach to portable ontology specifications, Knowl. Acquis. 5 (2) (1993) 199–220, http://dx.doi.org/10.1006/knac.1993.1008, URL https://www.sciencedirect.com/science/article/pii/S1042814383710083.

[2] O. Labbani Narsis, C. Nicolle, Cooperative knowledge elicitation for formal ontology design: an exploratory study applied in industry for knowledge management, in: 2023 IEEE International Conference on Systems, Man, and Cybernetics, SMC, 2023, pp. 3744–3749, http://dx.doi.org/10.1109/SMC53992.2023.10394180.

[3] V. Lytvyn, Y. Burov, V. Vysotska, O. Brodyak, Approach to automatic construction of interpretation functions during ontology learning, in: 2020 IEEE 15th International Conference on Computer Sciences and Information Technologies, CSIT, IEEE, Zbarazh, Ukraine, 2020, pp. 267–271, http://dx.doi.org/10.1109/CSIT49958.2020.9321920, URL https://ieeexplore.ieee.org/document/9321920/.

[4] C. Wang, X. Zhuo, P. Li, N. Chen, W. Wang, Z. Chen, An ontology-based framework for integrating remote sensing imagery, image products, and in situ observations, in: D. Palumbo (Ed.), J. Sensors 2020 (2020) 1–12, http://dx.doi.org/10.1155/2020/6912820, URL https://www.hindawi.com/journals/js/2020/6912820/.

[5] P. Cimiano, Ontology Learning and Population from Text: Algorithms, Evaluation and Applications, first ed., Springer US, Berlin, 2006.

[6] M.N. Asim, M. Wasim, M.U.G. Khan, W. Mahmood, H.M. Abbasi, A survey of ontology learning techniques and applications, Database 2018 (2018) bay101, http://dx.doi.org/10.1093/database/bay101.

[7] A.C. Khadir, H. Aliane, A. Guessoum, Ontology learning: Grand tour and challenges, Comp. Sci. Rev. 39 (2021) 100339, http://dx.doi.org/10.1016/j.cosrev.2020.100339, URL https://www.sciencedirect.com/science/article/pii/S1574013720304391.

[8] F.N. Al-Aswadi, H.Y. Chan, K.H. Gan, Automatic ontology construction from text: a review from shallow to deep learning trend, Artif. Intell. Rev. 53 (6) (2020) 3901–3928, http://dx.doi.org/10.1007/s10462-019-09782-9.

[9] R. Mosca, M. De Santo, R. Gaeta, Ontology learning from relational database: a review, J. Ambient Intell. Humaniz. Comput. 14 (12) (2023) 16841–16851, http://dx.doi.org/10.1007/s12652-023-04693-8.

[10] M.J. Somodevilla, D. Vilariño Ayala, I. Pineda, An overview on ontology learning tasks, Comput. Sist. 22 (1) (2018) http://dx.doi.org/10.13053/cys-22-1-2790, Number: 1, URL https://www.cys.cic.ipn.mx/ojs/index.php/CyS/article/view/2790.

[11] W. Wong, W. Liu, M. Bennamoun, Ontology learning from text: A look back and into the future, ACM Comput. Surv. 44 (4) (2012) 1–36, http://dx.doi.org/10.1145/2333112.2333115, URL https://dl.acm.org/doi/10.1145/2333112.2333115.

[12] R. Lourdusamy, S. Abraham, A Survey on Methods of Ontology Learning from Text, vol. 9, Springer International Publishing, Cham, 2020, pp. 113–123, http://dx.doi.org/10.1007/978-3-030-38501-9_11, Book Title: Intelligent Computing Paradigm and Cutting-edge Technologies Series Title: Learning and Analytics in Intelligent Systems, URL http://link.springer.com/10.1007/978-3-030-38501-9_11,

[13] S. Yuan, J. He, M. Wang, H. Zhou, Y. Ren, A review for ontology construction from unstructured texts by using deep learning, IoTML 2021, in: International Conference on Internet of Things and Machine Learning, vol. 12174, SPIE, 2022, pp. 315–322, http://dx.doi.org/10.1117/12.2628713, URL https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12174/121741D/A-review-for-ontology-construction-from-unstructured-texts-by-using/10.1117/12.2628713.full.

[14] Z.Z. Zulkipli, R. Maskat, N.H.I. Teo, A systematic literature review of automatic ontology construction, Indones. J. Electr. Eng. Comput. Sci. 28 (2) (2022) 878–889, http://dx.doi.org/10.11591/ijeecs.v28.i2.pp878-889, Number: 2, URL https://ijeecs.iaescore.com/index.php/IJEECS/article/view/28014.

[15] M. Hacherouf, S.N. Bahloul, C. Cruz, Transforming XML documents to OWL ontologies: A survey, J. Inf. Sci. 41 (2) (2015) 242–259, http://dx.doi.org/10.1177/0165551514565972, URL http://journals.sagepub.com/doi/10.1177/0165551514565972.

[16] P. Mayadewi, B. Sitohang, F.N. Azizah, Study relational database transformation to ontology, in: 2018 19th IEEE/ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing, SNPD, 2018, pp. 358–362, http://dx.doi.org/10.1109/SNPD.2018.8441102.

[17] B. Lakzaei, M. Shamsfard, Ontology learning from relational databases, Inform. Sci. 577 (2021) 280–297, http://dx.doi.org/10.1016/j.ins.2021.06.074, URL https://www.sciencedirect.com/science/article/pii/S0020025521006654.

[18] C. Ma, B. Molnár, Ontology learning from relational database: opportunities for semantic information integration, Vietnam J. Comput. Sci. 09 (01) (2022) 31–57, http://dx.doi.org/10.1142/S219688882150024X, URL https://www.worldscientific.com/doi/abs/10.1142/S219688882150024X.

[19] Q. Ji, G. Qi, H. Gao, T. Wu, Survey on schema induction from knowledge graphs, in: J. Zhao, F.v. Harmelen, J. Tang, X. Han, Q. Wang, X. Li (Eds.), Knowledge Graph and Semantic Computing. Knowledge Computing and Language Understanding, in: Communications in Computer and Information Science, Springer, Singapore, 2019, pp. 136–142, http://dx.doi.org/10.1007/978-981-13-3146-6_12.

[20] A. Konys, Knowledge repository of ontology learning tools from text, Procedia Comput. Sci. 159 (2019) 1614–1628, http://dx.doi.org/10.1016/j.procs.2019.09.332, URL https://www.sciencedirect.com/science/article/pii/S1877050919315339.

[21] J.a. Wątróbski, Ontology learning methods from text - an extensive knowledge-based approach, Procedia Comput. Sci. 176 (2020) 3356–3368, http://dx.doi.org/10.1016/j.procs.2020.09.061, URL https://www.sciencedirect.com/science/article/pii/S1877050920319566.

[22] T. Zengeya, J. Vincent Fonou-Dombeu, A review of state of the art deep learning models for ontology construction, IEEE Access 12 (2024) 82354–82383, http://dx.doi.org/10.1109/ACCESS.2024.3406426.

[23] B. Kitchenham, R. Pretorius, D. Budgen, O.P. Brereton, M. Turner, M. Niazi, S. Linkman, Systematic literature reviews in software engineering–a tertiary study, Inf. Softw. Technol. 52 (8) (2010) 792–805.

[24] M. Gusenbauer, N.R. Haddaway, Which academic search systems are suitable for systematic reviews or meta-analyses? Evaluating retrieval qualities of google scholar, PubMed, and 26 other resources, Res. Synth. Methods 11 (2) (2020) 181–217.

[25] J. Grosman, Findpapers: A tool for helping researchers who are looking for related works, 2020, https://github.com/jonatasgrosman/findpapers.

[26] G. Petrucci, M. Rospocher, C. Ghidini, Expressive ontology learning as neural machine translation, J. Web Semant. 52–53 (2018) 66–82, http://dx.doi.org/10.1016/j.websem.2018.10.002, URL https://linkinghub.elsevier.com/retrieve/pii/S1570826818300507.

[27] J.L. Austin, How to do things with words, second ed., in: The William James lectures, (1955) Clarendon Press, Oxford [Eng.], 1975.

[28] S. Sbissi, M. Mahfoudh, S. Gattoufi, A medical decision support system for cardiovacsular disease based on ontology learning, in: 2020 International Multi-Conference on: "Organization of Knowledge and Advanced Technologies", OCTA, IEEE, Tunis, Tunisia, 2020, pp. 1–9, http://dx.doi.org/10.1109/OCTA49274.2020.9151845, URL https://ieeexplore.ieee.org/document/9151845/.

[29] A.B. Rios-Alvarado, I. Lopez-Arevalo, E. Tello-Leal, V.J. Sosa-Sosa, An approach for learning expressive ontologies in medical domain, J. Med. Syst. 39 (8) (2015) 75, http://dx.doi.org/10.1007/s10916-015-0261-z.

[30] W. Zaouga, L.B.A. Rabai, A decision support system based on ontology learning from PMI' project risk management, in: A. Abraham, V. Piuri, N. Gandhi, P. Siarry, A. Kaklauskas, A. Madureira (Eds.), Intelligent Systems Design and Applications, in: Advances in Intelligent Systems and Computing, Springer International Publishing, Cham, 2021, pp. 732–742, http://dx.doi.org/10.1007/978-3-030-71187-0_67.

[31] B. Gyawali, A. Shimorina, C. Gardent, S. Cruz-Lara, M. Mahfoudh, Mapping natural language to description logic, in: E. Blomqvist, D. Maynard, A. Gangemi, R. Hoekstra, P. Hitzler, O. Hartig (Eds.), in: The Semantic Web, vol. 10249, Springer International Publishing, Cham, 2017, pp. 273–288, http://dx.doi.org/10.1007/978-3-319-58068-5_17, Series Title: Lecture Notes in Computer Science, URL https://link.springer.com/10.1007/978-3-319-58068-5_17.

[32] Z. Hu, W. Hou, X. Liu, Deep learning for named entity recognition: a survey, Neural Comput. Appl. (2024) 1–28.

[33] J.L. Martinez-Rodriguez, A. Hogan, I. Lopez-Arevalo, Information extraction meets the semantic web: a survey, Semant. Web 11 (2) (2020) 255–335.

[34] L. Wu, E. Sallinger, E. Sherkhonov, S. Vahdati, G. Gottlob, Rule Learning over Knowledge Graphs with Genetic Logic Programming, IEEE Computer Society, 2022, pp. 3373–3385, http://dx.doi.org/10.1109/ICDE53745.2022.00318, URL https://www.computer.org/csdl/proceedings-article/icde/2022/088300d373/1FwFh3oYeQM.

[35] P. Mateiu, A. Groza, C. Nica, Learning ontologies with relational concept analysis, in: 2022 IEEE 20th Jubilee World Symposium on Applied Machine Intelligence and Informatics, SAMI, 2022, pp. 000219–000224, http://dx.doi.org/10.1109/SAMI54271.2022.9780837.

[36] J. Potoniec, Inductive learning of OWL 2 property chains, IEEE Access 10 (2022) 25327–25340, http://dx.doi.org/10.1109/ACCESS.2022.3155816, URL https://ieeexplore.ieee.org/document/9724179/.

[37] R. Felin, A.G.B. Tettamanzi, Using grammar-based genetic programming for mining subsumption axioms involving complex class expressions, in: IEEE/WIC/ACM International Conference on Web Intelligence, ACM, ESSENDON VIC Australia, 2021, pp. 234–240, http://dx.doi.org/10.1145/3486622.3494025, URL https://dl.acm.org/doi/10.1145/3486622.3494025.

[38] F. Martel, A. Zouaq, Taxonomy Extraction Using Knowledge Graph Embeddings and Hierarchical Clustering, ACM, 2021, pp. 836–844, http://dx.doi.org/10.1145/3412841.3441959.

[39] J. Potoniec, Mining cardinality restrictions in OWL, Found. Comput. Decision Sci. 45 (3) (2020) 195–216, http://dx.doi.org/10.2478/fcds-2020-0011, URL https://www.sciendo.com/article/10.2478/fcds-2020-0011.

[40] J. Potoniec, Learning OWL 2 property characteristics as an explanation for an RNN, Bull. Pol. Acad. Sci. Tech. Sci. 68 (nr 6) (2020) http://dx.doi.org/10.24425/bpasts.2020.134625, URL http://yadda.icm.edu.pl/baztech/element/bwmeta1.element.baztech-3dd75bcb-7b63-4cf5-85ca-14a6a10d231b.

[41] T.H. Nguyen, A.G.B. Tettamanzi, Grammatical evolution to mine OWL disjointness axioms involving complex concept expressions, in: 2020 IEEE Congress on Evolutionary Computation, CEC, IEEE, Glasgow, United Kingdom, 2020, pp. 1–8, http://dx.doi.org/10.1109/CEC48606.2020.9185681, URL https://ieeexplore.ieee.org/document/9185681/.

[42] D. Malchiodi, C. Da Costa Pereira, A.G.B. Tettamanzi, Classifying candidate axioms via dimensionality reduction techniques, in: V.c. Torra, Y. Narukawa, J. Nin, N. Agell (Eds.), in: Modeling Decisions for Artificial Intelligence, vol. 12256, Springer International Publishing, Cham, 2020, pp. 179–191, http://dx.doi.org/10.1007/978-3-030-57524-3_15, Series Title: Lecture Notes in Computer Science, URL https://link.springer.com/10.1007/978-3-030-57524-3_15.

[43] G. Rizzo, N. Fanizzi, C. d'Amato, Class expression induction as concept space exploration: From DL-Foil to DL-Focl, Future Gener. Comput. Syst. 108 (2020) 256–272, http://dx.doi.org/10.1016/j.future.2020.02.071, URL https://www.sciencedirect.com/science/article/pii/S0167739X19303991.

[44] T.H. Nguyen, A.G.B. Tettamanzi, A multi-objective evolutionary approach to class disjointness axiom discovery, in: 2020 IEEE/WIC/ACM International Joint Conference on Web Intelligence and Intelligent Agent Technology, WI-IAT, IEEE, Melbourne, Australia, 2020, pp. 275–282, http://dx.doi.org/10.1109/WIIAT50758.2020.00040, URL https://ieeexplore.ieee.org/document/9457766/.

[45] T.H. Nguyen, A.G.B. Tettamanzi, Learning class disjointness axioms using grammatical evolution, in: L. Sekanina, T. Hu, N. Louren¸co, H. Richter, P. Garcí a Sánchez (Eds.), in: Genetic Programming, vol. 11451, Springer International Publishing, Cham, 2019, pp. 278–294, http://dx.doi.org/10.1007/978-3-030-16670-0_18, Series Title: Lecture Notes in Computer Science, URL http://link.springer.com/10.1007/978-3-030-16670-0_18.

[46] L. Palacios Medinacelli, Y. Ma, C. Reynaud, G. Lortal, Knowledge based situation discovery for avionics maintenance, in: Proceedings of the 10th International Conference on Knowledge Capture, ACM, Marina Del Rey CA USA, 2019, pp. 155–162, http://dx.doi.org/10.1145/3360901.3364430, URL https://dl.acm.org/doi/10.1145/3360901.3364430.

[47] T.H. Nguyen, A.G. Tettamanzi, An evolutionary approach to class disjointness axiom discovery, in: IEEE/WIC/ACM International Conference on Web Intelligence, ACM, Thessaloniki Greece, 2019, pp. 68–75, http://dx.doi.org/10.1145/3350546.3352502, URL https://dl.acm.org/doi/10.1145/3350546.3352502.

[48] S. Ortona, V.V. Meduri, P. Papotti, Robust discovery of positive and negative rules in knowledge bases, in: 2018 IEEE 34th International Conference on Data Engineering (ICDE), IEEE, Paris, 2018, pp. 1168–1179, http://dx.doi.org/10.1109/ICDE.2018.00108, URL https://ieeexplore.ieee.org/document/8509329/.

[49] I. Huitzil, U. Straccia, N. Díaz-Rodríguez, F. Bobillo, Datil: learning fuzzy ontology datatypes, in: J. Medina, M. Ojeda-Aciego, J.L. Verdegay, D.A. Pelta, I.P. Cabrera, B. Bouchon-Meunier, R.R. Yager (Eds.), Information Processing and Management of Uncertainty in Knowledge-Based Systems. Theory and Foundations, 854, Springer International Publishing, Cham, 2018, pp. 100–112, http://dx.doi.org/10.1007/978-3-319-91476-3_9, Series Title: Communications in Computer and Information Science, URL http://link.springer.com/10.1007/978-3-319-91476-3_9.

[50] J. Potoniec, P. Jakubowski, A. Lawrynowicz, Swift Linked Data Miner: Mining OWL 2 EL class expressions directly from online RDF datasets, J. Web Semant. 46–47 (2017) 31–50, http://dx.doi.org/10.1016/j.websem.2017.08.001, URL https://linkinghub.elsevier.com/retrieve/pii/S157082681730032X.

[51] R. Irny, P. Sreenivasa Kumar, Mining inverse and symmetric axioms in linked data, in: Z. Wang, A.-Y. Turhan, K. Wang, X. Zhang (Eds.), Semantic Technology, in: Lecture Notes in Computer Science, Springer International Publishing, Cham, 2017, pp. 215–231, http://dx.doi.org/10.1007/978-3-319-70682-5_14.

[52] C. d'Amato, S. Staab, A.G.B. Tettamanzi, T.D. Minh, F. Gandon, Ontology enrichment by discovering multi-relational association rules from ontological knowledge bases, in: Proceedings of the 31st Annual ACM Symposium on Applied Computing, ACM, Pisa Italy, 2016, pp. 333–338, http://dx.doi.org/10.1145/2851613.2851842, URL https://dl.acm.org/doi/10.1145/2851613.2851842.

[53] B.T. Luong, S. Ruggieri, F. Turini, Classification rule mining supported by ontology for discrimination discovery, in: 2016 IEEE 16th International Conference on Data Mining Workshops, ICDMW, IEEE, Barcelona, Spain, 2016, pp. 868–875, http://dx.doi.org/10.1109/ICDMW.2016.0128, URL http://ieeexplore.ieee.org/document/7836759/.

[54] M. Zhu, Z. Gao, J.Z. Pan, Y. Zhao, Y. Xu, Z. Quan, TBox learning from incomplete data by inference in BelNet+, Knowl.-Based Syst. 75 (2015) 30–40, http://dx.doi.org/10.1016/j.knosys.2014.11.004, URL https://linkinghub.elsevier.com/retrieve/pii/S0950705114003967.

[55] J. Völker, D. Fleischhacker, H. Stuckenschmidt, Automatic acquisition of class disjointness, J. Web Semant. 35 (2015) 124–139, http://dx.doi.org/10.1016/j.websem.2015.07.001, URL https://www.sciencedirect.com/science/article/pii/S157082681500061X.

[56] Z. Abedjan, T. Gruetze, A. Jentzsch, F. Naumann, Profiling and mining RDF data with ProLOD++, in: 2014 IEEE 30th International Conference on Data Engineering, IEEE, Chicago, IL, USA, 2014, pp. 1198–1201, http://dx.doi.org/10.1109/ICDE.2014.6816740, URL http://ieeexplore.ieee.org/document/6816740/.

[57] E. Rafael Hruschka, R. Garcia Leonel Miani, Analyzing the use of obvious and generalized association rules in a large knowledge base, in: 2014 14th International Conference on Hybrid Intelligent Systems, IEEE, Kuwait, Kuwait, 2014, pp. 1–6, http://dx.doi.org/10.1109/HIS.2014.7086179, URL https://ieeexplore.ieee.org/document/7086179/.

[58] D. Borchmann, Towards an error-tolerant construction of $\mathcal{EL}^\bot$-ontologies from data using formal concept analysis, in: D. Hutchison, T. Kanade, J. Kittler, J.M. Kleinberg, F. Mattern, J.C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M.Y. Vardi, G. Weikum, P. Cellier, F. Distel, B. Ganter (Eds.), in: Formal Concept Analysis, vol. 7880, Springer Berlin Heidelberg, Berlin, Heidelberg, 2013, pp. 60–75, http://dx.doi.org/10.1007/978-3-642-38317-5_4, Series Title: Lecture Notes in Computer Science, URL http://link.springer.com/10.1007/978-3-642-38317-5_4.

[59] M. Zhu, Z. Gao, J.Z. Pan, Y. Zhao, Y. Xu, Z. Quan, Ontology learning from incomplete semantic web data by BelNet, in: 2013 IEEE 25th International Conference on Tools with Artificial Intelligence, IEEE, Herndon, VA, USA, 2013, pp. 761–768, http://dx.doi.org/10.1109/ICTAI.2013.117, URL http://ieeexplore.ieee.org/document/6735328/.

[60] D. Fleischhacker, J. Völker, H. Stuckenschmidt, Mining RDF data for property axioms, in: D. Hutchison, T. Kanade, J. Kittler, J.M. Kleinberg, F. Mattern, J.C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M.Y. Vardi, G. Weikum, R. Meersman, H. Panetto, T. Dillon, S. Rinderle-Ma, P. Dadam, X. Zhou, S. Pearson, A. Ferscha, S. Bergamaschi, I.F. Cruz (Eds.), OTM 2012, in: On the Move to Meaningful Internet Systems, vol. 7566, Springer Berlin Heidelberg, Berlin, Heidelberg, 2012, pp. 718–735, http://dx.doi.org/10.1007/978-3-642-33615-7_18, Series Title: Lecture Notes in Computer Science, URL http://link.springer.com/10.1007/978-3-642-33615-7_18.

[61] B. Lakzaei, M. Shamsfard, Ontology learning from relational databases, Inform. Sci. 577 (2021) 280–297, http://dx.doi.org/10.1016/j.ins.2021.06.074, URL https://www.sciencedirect.com/science/article/pii/S0020025521006654.

[62] C. Ma, B. Molnár, A. Benczúr, A semi-automatic semantic consistency-checking method for learning ontology from relational database, Information 12 (5) (2021) 188, http://dx.doi.org/10.3390/info12050188, URL https://www.mdpi.com/2078-2489/12/5/188.

[63] N.K. Soe, T.T. Yee, E.C. Htoon, C2Ont: an OWL ontology learning approach from apache cassandra, in: 2019 International Conference on Advanced Information Technologies, ICAIT, 2019, pp. 31–36, http://dx.doi.org/10.1109/AITC.2019.8921025.

[64] H. Gao, G. Qi, Q. Ji, Schema induction from incomplete semantic data, Intell. Data Anal. 22 (6) (2018) 1337–1353, http://dx.doi.org/10.3233/IDA-173514, URL https://www.medra.org/servlet/aliasResolver?alias=iospress&doi=10.3233/IDA-173514.

[65] C.-h. Liao, Y.-f. Wu, G.-h. King, Research on learning OWL ontology from relational database, J. Phys. Conf. Ser. 1176 (2) (2019) 022031, http://dx.doi.org/10.1088/1742-6596/1176/2/022031, Publisher: IOP Publishing.

[66] P. McBrien, L. Al Khuzayem, OWLRel: learning rich ontologies from relational databases, 2016, 482 Accepted: 2017-06-21T07:48:27Z Publisher: Vilnius University, University of Latvia, Latvia University of Agriculture, Institute of Mathematics and Informatics of University of Latvia, URL http://spiral.imperial.ac.uk/handle/10044/1/47878.

[67] B. El Idrissi, S. Baï na, K. Baï na, Bringing flexibility to ontology learning from relational databases, in: 2015 IEEE/ACS 12th International Conference of Computer Systems and Applications, AICCSA, 2015, pp. 1–8, http://dx.doi.org/10.1109/AICCSA.2015.7507213, ISSN: 2161-5330.

[68] Y. Lu, C. Li, R. Sun, L. Wu, G. Zhu, Improved automatic conversion rules and algorithm from relational schema to ontology, J. Comput. Theor. Nanosci. 11 (6) (2014) 1537–1541, http://dx.doi.org/10.1166/jctn.2014.3532, URL http://openurl.ingenta.com/content/xref?genre=article&issn=1546-1955&volume=11&issue=6&spage=1537.

[69] F. Zhang, Z. Ma, L. Yan, Construction of fuzzy ontologies from fuzzy XML models, Knowl.-Based Syst. 42 (2013) 20–39, http://dx.doi.org/10.1016/j.knosys.2012.12.015, URL https://linkinghub.elsevier.com/retrieve/pii/S0950705113000063.

[70] C.B. Abhilash, K. Mahesh, Ontology-based method for semantic association rules, in: 2022 IEEE 19th India Council International Conference, INDICON, IEEE, Kochi, India, 2022, pp. 1–7, http://dx.doi.org/10.1109/INDICON56171.2022.10039950, URL https://ieeexplore.ieee.org/document/10039950/.

[71] A.C.F. Petri, D.F. Silva, Towards logical association rule mining on ontology-based semantic trajectories, in: 2020 19th IEEE International Conference on Machine Learning and Applications, ICMLA, IEEE, Miami, FL, USA, 2020, pp. 586–591, http://dx.doi.org/10.1109/ICMLA51294.2020.00098, URL https://ieeexplore.ieee.org/document/9356250/.

[72] M. Chang, G. D'Aniello, M. Gaeta, F. Orciuoli, D. Sampson, C. Simonelli, Building ontology-driven tutoring models for intelligent tutoring systems using data mining, IEEE Access 8 (2020) 48151–48162, http://dx.doi.org/10.1109/ACCESS.2020.2979281, URL https://ieeexplore.ieee.org/document/9031710/.

[73] A.B. Rios-Alvarado, J.L. Martinez-Rodriguez, T.Y. Guerrero-Melendez, A.J. Rodriguez-Rodriguez, D.T. Vargas-Requena, Axiomatic relation extraction from text in the domain of tourism, in: B. Villazón-Terrazas, F. Ortiz-Rodrí guez, S.M. Tiwari, S.K. Shandilya (Eds.), Knowledge Graphs and Semantic Web, in: Communications in Computer and Information Science, Springer International Publishing, Cham, 2020, pp. 14–28, http://dx.doi.org/10.1007/978-3-030-65384-2_2.

[74] C. Kacfah Emani, C. Ferreira Da Silva, B. Fiès, P. Ghodous, NALDO: From natural language definitions to OWL expressions, Data Knowl. Eng. 122 (2019) 130–141, http://dx.doi.org/10.1016/j.datak.2019.06.002, URL https://linkinghub.elsevier.com/retrieve/pii/S0169023X18306086.

[75] A. Jiomekong, G. Camara, M. Tchuente, Extracting ontological knowledge from Java source code using Hidden Markov Models, Open Comput. Sci. 9 (1) (2019) 181–199, http://dx.doi.org/10.1515/comp-2019-0013, Publisher: De Gruyter Open Access, URL https://www.degruyter.com/document/doi/10.1515/comp-2019-0013/html.

[76] S. Dami, A.A. Barforoush, H. Shirazi, News events prediction using Markov logic networks, J. Inf. Sci. 44 (1) (2018) 91–109, http://dx.doi.org/10.1177/0165551516673285, URL http://journals.sagepub.com/doi/10.1177/0165551516673285.

[77] K.A. Mathews, P.S. Kumar, Extracting ontological knowledge from textual descriptions through grammar-based transformation, in: Proceedings of the Knowledge Capture Conference, ACM, Austin TX USA, 2017, pp. 1–4, http://dx.doi.org/10.1145/3148011.3148034, URL https://dl.acm.org/doi/10.1145/3148011.3148034.

[78] S. Saad, B. Idrus, Axiomatic ontology learning approaches for english translation of the meaning of quranic texts, MATEC Web Conf. 135 (2017) 00075, http://dx.doi.org/10.1051/matecconf/201713500075, URL http://www.matec-conferences.org/10.1051/matecconf/201713500075.

[79] R.R.D. Azevedo, F. Freitas, R. Rocha, J.A.A.D. Menezes, C.M.D.O. Rodrigues, M.C. Gomes, Representing knowledge in DL ALC from text, Procedia Comput. Sci. 35 (2014) 176–185, http://dx.doi.org/10.1016/j.procs.2014.08.097, URL https://linkinghub.elsevier.com/retrieve/pii/S187705091401062X.

[80] N. Sachdeva, M. Jain, R. Mutharaju, Extraction of union and intersection axioms from biomedical text, in: R. Verborgh, A. Dimou, A. Hogan, C. d'Amato, I. Tiddi, A. Bröring, S. Mayer, F. Ongenae, R. Tommasini, M. Alam (Eds.), in: The Semantic Web: ESWC 2021 Satellite Events, vol. 12739, Springer International Publishing, Cham, 2021, pp. 147–151, http://dx.doi.org/10.1007/978-3-030-80418-3_27, Series Title: Lecture Notes in Computer Science, URL https://link.springer.com/10.1007/978-3-030-80418-3_27.

[81] D. Sánchez, A. Moreno, L. Del Vasto-Terrientes, Learning relation axioms from text: An automatic Web-based approach, Expert Syst. Appl. 39 (5) (2012) 5792–5805, http://dx.doi.org/10.1016/j.eswa.2011.11.088, URL https://linkinghub.elsevier.com/retrieve/pii/S0957417411016344.

[82] K. Yordanova, From textual instructions to sensor-based recognition of user behaviour, in: Companion Publication of the 21st International Conference on Intelligent User Interfaces, ACM, Sonoma California USA, 2016, pp. 67–73, http://dx.doi.org/10.1145/2876456.2879488, URL https://dl.acm.org/doi/10.1145/2876456.2879488.

[83] D. Malchiodi, A.G.B. Tettamanzi, Predicting the possibilistic score of OWL axioms through modified support vector clustering, in: Proceedings of the 33rd Annual ACM Symposium on Applied Computing, ACM, Pau France, 2018, pp. 1984–1991, http://dx.doi.org/10.1145/3167132.3167345, URL https://dl.acm.org/doi/10.1145/3167132.3167345.

[84] P. Buitelaar, P. Cimiano, B. Magnini, Ontology Learning from Text: An Overview, 2005, p. 12.

[85] S. Ghidalia, O.L. Narsis, A. Bertaux, C. Nicolle, Combining machine learning and ontology: a systematic literature review, 2024, http://dx.doi.org/10.48550/arXiv.2401.07744, arXiv:2401.07744 [cs] version: 1, URL http://arxiv.org/abs/2401.07744.

[86] L. Bühmann, J. Lehmann, P. Westphal, S. Bin, DL-learner structured machine learning on semantic web data, in: Companion of the the Web Conference 2018 on the Web Conference 2018, WWW '18, ACM Press, Lyon, France, 2018, pp. 467–471, http://dx.doi.org/10.1145/3184558.3186235, URL http://dl.acm.org/citation.cfm?doid=3184558.3186235.

[87] L. Galárraga, C. Teflioudi, K. Hose, F.M. Suchanek, Fast rule mining in ontological knowledge bases with AMIE$$+$$, VLDB J. 24 (6) (2015) 707–730, http://dx.doi.org/10.1007/s00778-015-0394-1.

[88] A. Ozaki, Learning description logic ontologies: five approaches. where do they stand? KI - Künstliche Intell. 34 (3) (2020) 317–327, http://dx.doi.org/10.1007/s13218-020-00656-9, URL https://link.springer.com/10.1007/s13218-020-00656-9.

[89] M. Schaeffer, M. Sesboüé, J.-P. Kotowicz, N. Delestre, C. Zanni-Merk, OLAF: an ontology learning applied framework, Procedia Comput. Sci. 225 (2023) 2106–2115, http://dx.doi.org/10.1016/j.procs.2023.10.201, URL https://www.sciencedirect.com/science/article/pii/S1877050923013595.

[90] A. Ballout, C. da Costa Pereira, A.G.B. Tettamanzi, Predicting the acceptability of atomic candidate OWL class axioms, in: 2023 IEEE/WIC International Conference on Web Intelligence and Intelligent Agent Technology, WI-IAT, 2023, pp. 339–345, http://dx.doi.org/10.1109/WI-IAT59888.2023.00055, URL https://ieeexplore.ieee.org/document/10350129.

[91] M. Grüninger, A. Chow, J. Wong, Semiautomatic design of ontologies, in: J.a.P.A. Almeida, M. Kaczmarek-Heß, A. Koschmider, H.A. Proper (Eds.), in: The Practice of Enterprise Modeling, vol. 497, Springer Nature Switzerland, Cham, 2024, pp. 143–158, http://dx.doi.org/10.1007/978-3-031-48583-1_9, Series Title: Lecture Notes in Business Information Processing, URL https://link.springer.com/10.1007/978-3-031-48583-1_9.